Online Appendix

What Connects Local Housing Markets? Regulation, Migration and the National Cycle

Oğuz Yurtoğlu*

October 1, 2026

This Online Appendix collects the robustness checks, complete tables and descriptive exhibits behind the paper. Sections, tables and figures here carry the prefix OA; a reference without it points to the paper or its printed appendix. Where an exhibit has a factor-adjusted version, it sits beside the raw one in the same table. Units and inference follow the paper (Section 5.4), and every table note states its own.

OA.1 Networks and Connectedness Indices

OA.1.1 Monthly house-price growth

Figure OA.1 draws every state's monthly growth series, the input to the connectedness system. The extreme movements of 1981–82 described in Appendix A stand out against the rest of the sample.

OA.1.2 Testing the decline in the distance gradient

The per-endpoint regressions behind Figure 6 each carry their own covariance matrix, so the fall in the gradient cannot be tested from them. The same regressions fitted jointly—log distance interacted with endpoint, with source-by-endpoint and target-by-endpoint effects—leave every point estimate numerically unchanged, because fully interacted node effects make the Poisson likelihood separable by endpoint. At twelve endpoints the two sets of estimates agree to machine precision. What the joint fit adds is one covariance matrix, and with it a Wald test. The mean gradient over 2015–2025 is 0.301 below the mean over 1990–1999, with a two-way clustered standard error of 0.045. Two cautions attach. The clustering has 48 source and 48 target states, so the effective number of clusters is small relative to the 252 interacted coefficients, and the joint covariance is rank-deficient by construction; the contrast tested here is one-dimensional and needs only its own variance, but a joint test across the coefficients would not be supported.

A second question about the same regressions does not resolve as expected. The figure uses the row-normalized cell, and it is natural to suppose that the gradient would be identical on

*Koç University, Department of Economics. Email: oyurtoglu22@ku.edu.tr

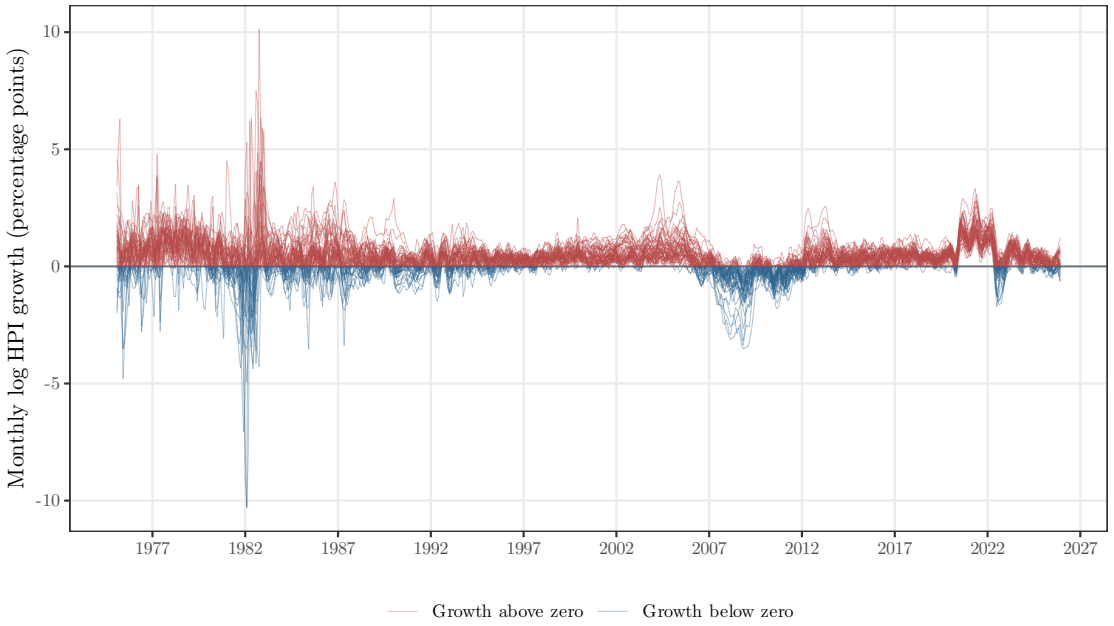


Figure OA.1: Monthly HPI growth across the 48 contiguous states, 1975–2025

Note: Every line is one state series used in the connectedness system. Growth is multiplied by 100 and reported in log percentage points.

unnormalized cells, since the normalization divides a target's row at an endpoint by a quantity that is constant within target-endpoint and a log link with target-by-endpoint effects absorbs constants exactly. That reasoning holds for a constant common to every row, which leaves the gradients unchanged to 10^{-12}, and fails for one that varies across rows, which the row total does. Poisson quasi-likelihood sets the group effect so that fitted and observed totals agree within the group, and a row rescaled by c_g then contributes c_g times as much to the slope score. The fixed effect absorbs the level, not the weight. Rescaling each target-endpoint row by its own constant moves the gradients materially. The gradient on unnormalized cells is therefore a different estimand from the gradient on normalized cells, and the retained outputs carry the normalized cell only.

The row weights can be equalized without the unnormalized cells. Figure OA.2 re-estimates every gradient of Figure 6 on the conditional shares of equation (12), which sum to 100 in every target row and so give every target the same weight. The conditional and unconditional gradients correlate at 0.999 in the raw state network, 0.996 in the factor-adjusted one, 1.000 in the raw metropolitan network and 0.999 in the factor-adjusted one, and no endpoint differs by more than 0.031. The raw state gradient still falls, from 0.408 in the 1990s to 0.118 during 2020–2025, while the factor-adjusted one averages 0.159 over the annual networks and has no trend distinguishable from zero. The contrast between the two networks does not come from how targets are weighted.

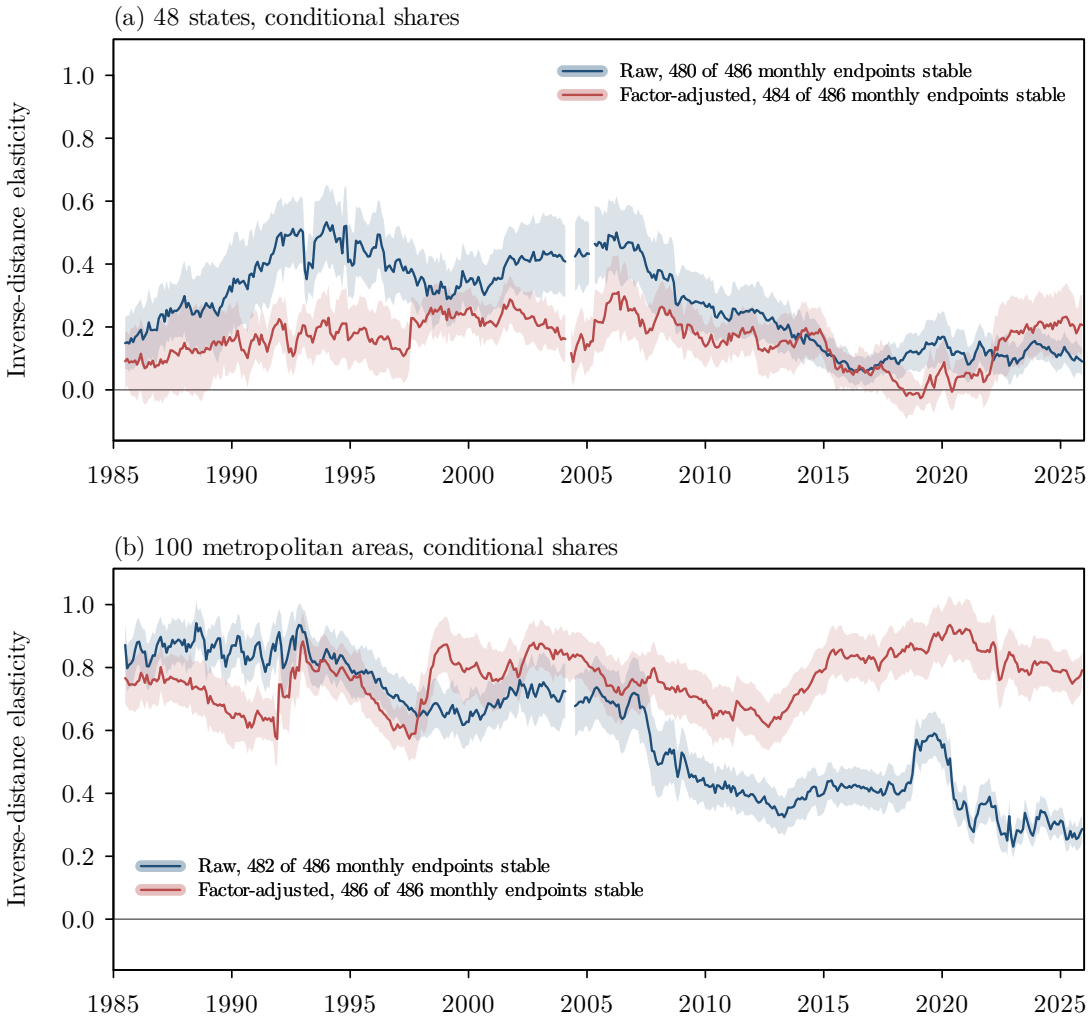


Figure OA.2: The distance gradient on conditional shares, raw and factor-adjusted, July 1985–December 2025

Note: Figure 6 re-estimated on the conditional share $\bar{C}_{ij,t} = 100C_{ij,t}/F_{i,t}$ of equation (12), which sums to 100 across the sources of every target and so gives every target row the same weight. Panel (a) covers the 48 states and 2,256 directed pairs at each monthly endpoint, and panel (b) the 100 metropolitan areas and 9,900 pairs. The raw network is navy and the factor-adjusted one red. Every regression includes source-node and target-node fixed effects, and the bands are 95-percent intervals with two-way source and target clustering. Endpoints without a variance decomposition are drawn as gaps, six for the raw state network, two for the factor-adjusted one, four for the raw metropolitan network and none for the factor-adjusted one.

OA.1.3 Metropolitan connectedness over the national cycle

Figure OA.3 repeats Figure 4 for the 100 metropolitan areas. The raw index rises from an average of 87.2 over the annual networks of 1985–1989 to 96.2 since 2020, and the factor-adjusted index averages 87.4 and 85.7 over the same endpoints, 86.2 across all monthly endpoints, and stays between 81.8 and 89.1. As at state scale, the two are weakly negatively correlated month by month, −0.33, so the rise of metropolitan connectedness also runs through the national factor. The factor-adjusted indexes of the two systems move together, with levels that correlate at 0.83 across the monthly endpoints.

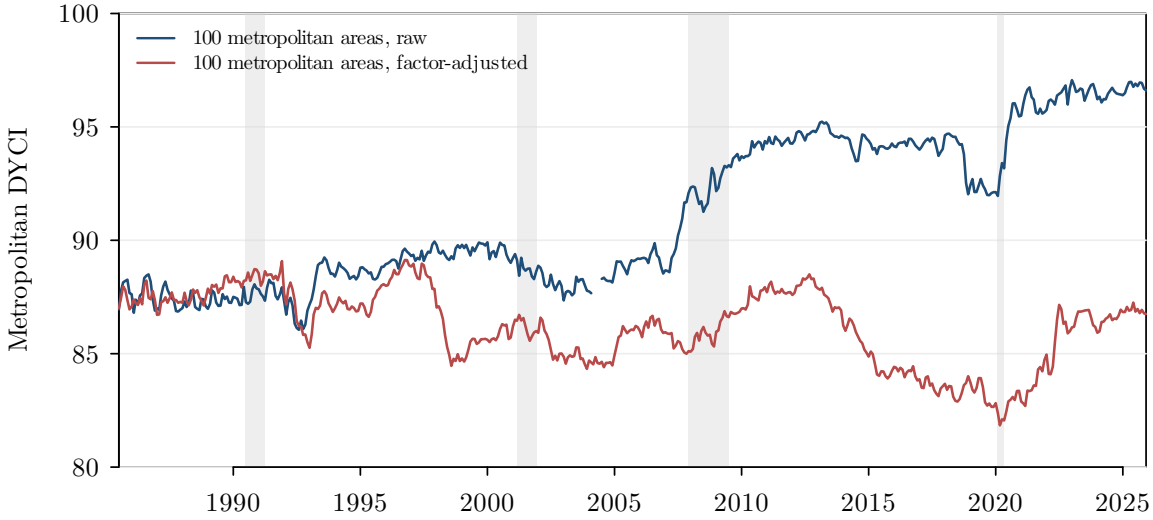


Figure OA.3: Metropolitan connectedness over the national housing cycle, raw and factor-adjusted, July 1985–December 2025

Note: Monthly Diebold–Yilmaz Connectedness Index of the 100-metropolitan-area system from the 120-month rolling elastic-net VAR(2) at the benchmark specification of Appendix D.2, in navy for the raw network and in red for the factor-adjusted one, estimated after each window's first principal component of the 100 areas' monthly growth is removed (Section 2.3). The four raw endpoints March–June 2004, whose fitted companion matrix has an eigenvalue outside the unit circle, have no variance decomposition and are drawn as gaps, and the factor-adjusted network is stable at all 486 endpoints. Levels are not comparable with those of the 48-state system (Section OA.1.4). Shaded areas denote NBER recessions.

OA.1.4 The two systems side by side

Figure OA.4 sets the state index beside the index for the 100-metropolitan-area system, and the comparison has to be read with care. The metropolitan index is both higher and much steadier. Over the 480 monthly endpoints from July 1985 to December 2025 at which both systems are dynamically stable, it averages 91.2 against 78.2, a gap that averages 13.0 points, and it moves between 86.1 and 97.1 while the state index moves between 65.1 and 93.5. These are monthly-endpoint averages. Over the 41 annual networks that the external-connectedness tables use the two average 91.2 and 78.1, and over the 21 from 2005 that the allocation tables use, 93.9 and 85.3. Part of the level gap

is mechanical. External connectedness is the share of a target's forecast-error variance assigned to other nodes, and a metropolitan target has 99 external sources where a state has 47, so comparable underlying dependence registers as a larger external share in the larger system. The difference in amplitude matters more. The standard deviation of the metropolitan index is 3.30 points against 8.58 for the state index, so the figure's right-hand axis is scaled by the ratio of the two standard deviations, and one metropolitan point occupies 2.60 state points. The panel is built to compare timing, not amplitude, and it should not be read as showing two indexes that move by similar amounts. What it does show is that the two scales turn together. Their levels correlate at 0.985 and their twelve-month changes at 0.793.

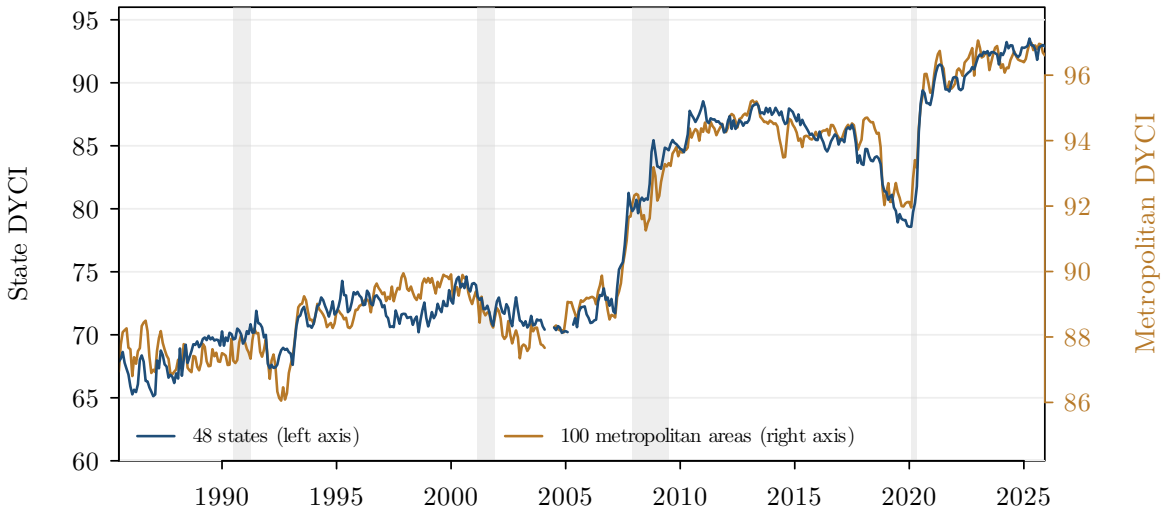


Figure OA.4: The state and metropolitan systems, each on its own scale, July 1985–December 2025

Note: The state index is read on the left axis and the index for the 100-metropolitan-area system on the right, both at the benchmark penalty rule and both estimated under hv-block cross-validation on the same 486 monthly endpoints. The right axis maps the metropolitan series through its own mean and standard deviation, so that one metropolitan point spans 2.60 state points. The figure is scaled so that the timing of the two series is comparable and their amplitudes are not. Endpoints whose fitted companion matrix has an eigenvalue outside the unit circle have no variance decomposition and are drawn as gaps, six for the benchmark state system and four for the metropolitan system. Shaded areas denote NBER recessions.

OA.1.5 Systemwide connectedness under the three own-lag treatments

Figure OA.5 draws the monthly index of the raw state system under the benchmark and the two own-lag alternatives of Section 7.1. The own-lags-free index moves with the benchmark, with levels that correlate at 0.985 and twelve-month changes at 0.77 over the 473 endpoints at which both exist. The eight-own-lag index has no variance decomposition at 110 of its 484 monthly endpoints, 98 of them in 2002–2012, and where it exists its levels correlate with the benchmark at 0.865. Its path through the boom and the financial crisis therefore cannot be compared with the benchmark's. Where all three indexes exist, they average 78.2, 71.6 and 65.1.

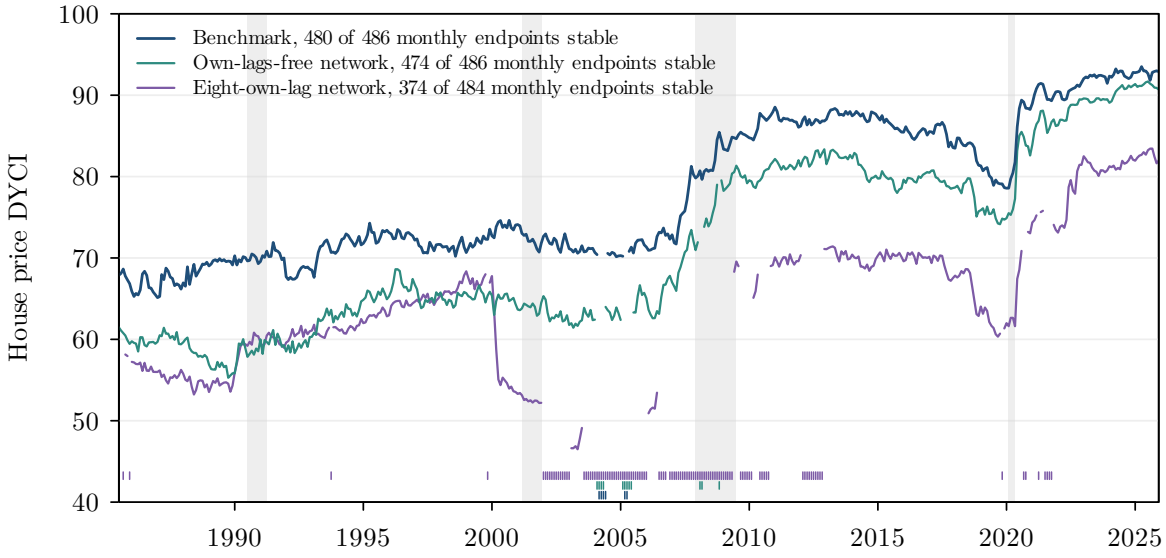


Figure OA.5: Systemwide connectedness under the three own-lag treatments, July 1985–December 2025

Note: Monthly Diebold–Yilmaz Connectedness Index of the raw 48-state system at the benchmark (navy), in the own-lags-free network (green) and in the eight-own-lag network (purple), all with hv-block penalty selection. Endpoints without a variance decomposition are drawn as gaps and marked by ticks along the bottom, one row per network with the benchmark lowest (6, 12 and 110 endpoints). The eight-own-lag series starts in September 1985 because the two earlier windows lack eight months of lags. Shaded areas denote NBER recessions.

OA.1.6 Lag order and the information criterion

Table OA.1 reports the Bayesian information criterion behind the lag order of Appendix B.1, computed on the benchmark estimator itself with one, two and three lags of every state. With the penalized fit's nonzero coefficients as degrees of freedom, two lags has the lowest criterion in 40 of the 41 annual networks. With each equation refitted by least squares on its retained slopes, one lag is still rejected by a wide margin, but three lags has a slightly lower criterion than two at 25 of the 38 endpoints at which every equation can be refitted, so on that count the criterion does not separate two lags from three. At three endpoints one three-lag equation retains more slopes than it has observations and cannot be refitted.

Table OA.1: Lag order and the Bayesian information criterion at the benchmark estimator, 48 states

Lags K	Active slopes per equation	Penalized fit			Post-selection refit	
		BIC	Difference from $K = 2$	Endpoints preferred	Difference from $K = 2$	Endpoints preferred
Panel A. The 41 annual networks 1985–2025 (38 in the post-selection columns)						
1	13.8	−70,337	+857	0	+2,005	0
2	25.1	−71,194	–	40	–	13
3	25.7	−70,764	+430	1	−118	25
Panel B. The 21 annual networks 2005–2025 (19 in the post-selection columns)						
1	14.5	−71,748	+921	0	+2,197	0
2	27.0	−72,669	–	21	–	6
3	28.2	−72,183	+486	0	−106	13

Note: The 48-state system. In each annual network the benchmark estimator of Section 2 is refitted with $K = 1$, 2 and 3 lags of every state and nothing else changed. Each equation is an elastic net with a mixing weight of 0.5 on the same 120 response months, with pre-window months supplying the lags, and its penalty is the one that minimizes hv-block validation loss over twelve contiguous ten-month blocks, with a buffer of K months on either side of each block. Own lags carry the cross-lag penalty. For equation e the criterion is $n \log(\mathrm{RSS}_e/n) + \mathrm{df}_e \log n$ with $n = 120$. In the penalized columns RSS_e is the in-sample residual sum of squares at the selected penalty and df_e is the number of nonzero slopes plus the intercept, the usual effective degrees of freedom of a lasso-type fit, exact for the lasso and an approximation for the elastic net. In the post-selection columns each equation is refitted by least squares on the slopes the penalty retains, with the same count. At three endpoints (1999, 2014 and 2021) one three-lag equation retains 120 or more slopes and cannot be refitted, so the post-selection columns use the endpoints at which every equation of every lag order can be. The system criterion is the sum over the 48 equations. BIC is its mean over the endpoints, and a difference is the mean of the endpoint-by-endpoint differences from $K = 2$, so a positive difference favors two lags. Endpoints preferred counts the endpoints at which a lag order has the lowest criterion. Active slopes are the nonzero lag coefficients per equation, own lags included. The $K = 2$ fits reproduce the stored benchmark network at every endpoint.

OA.1.7 The first-component share and migration exposure

If disturbances in regulated markets spread to the markets their migrants connect them to, those markets would load more heavily on the common component, and part of its growth would reflect propagation rather than common shocks. Table OA.2 regresses each state's first-component share, the object of Appendix Table C.2, on its migration exposure to regulated markets, $E_{i,t} = \sum_{j \neq i} \psi_{ij,t} R_j$ with R_j standardized WRLURI 2006, and on its own regulation, with year effects, over the 21 annual networks from 2005 for which exposure exists. Exposure is standardized across the 1,008 state-years. In the primary specification the exposure coefficient is 0.73 percentage points (1.82), with a wild bootstrap p-value of 0.70 and a 95-percent interval of $[-3.76, 4.56]$, while the state's own regulation carries 5.19 (1.93). With the controls of Table 2 added the exposure coefficient is 4.76 (1.86), with an interval of $[0.63, 9.04]$. With openness added it is 1.15, with exposure measured on the conditional shares −0.51, and with the state's first-component loading as the outcome 0.012, none distinguishable from zero. The diagnostic therefore gives no robust sign that the common component loads on exposure to regulated markets.

Table OA.2: The first-component share and migration exposure to regulated markets, 48 states, 2005–2025

Dependent variable	First-component share (percentage points)				First-component loading
	(1)	(2)	(3)	(4)	(5)
Exposure to regulated markets	0.73	4.76**	1.15	−0.51	0.012
	(1.82)	(1.86)	(2.10)	(1.99)	(0.014)
	[−3.76, 4.56]	[0.63, 9.04]	[−3.56, 5.93]	[−4.73, 3.85]	[−0.022, 0.041]
Own WRLURI 2006	5.19***	2.11	4.83**	6.11**	0.031**
	(1.93)	(1.81)	(2.06)	(2.12)	(0.015)
	[1.20, 9.99]	[−2.00, 6.81]	[0.40, 9.70]	[1.49, 10.79]	[0.000, 0.068]
Controls of Table 2	–	✓	–	–	–
Migration openness	–	–	✓	–	–
Exposure built from conditional shares	–	–	–	✓	–
Year fixed effects	✓	✓	✓	✓	✓
Mean of the dependent variable	63.60	63.60	63.60	63.60	0.78
State-years	1,008	1,008	1,008	1,008	1,008

Note: The 48 contiguous states in the 21 annual networks 2005–2025. The first-component share is $100R^2$ of the state's monthly log house-price growth on the first principal component of the 48 standardized growth series over the 120-month window ending at the endpoint, the object of Table C.2. Column (5) uses the state's first-component loading instead, the correlation of its standardized growth with that component, signed so that the 48 loadings have a positive mean. Exposure to regulated markets is $E_{i,t} = \sum_{j \neq i} \psi_{ij,t} R_j$, with $\psi_{ij,t}$ the migration exposure of equation 7 (IRS flows pooled over $t-15$ through $t-11$, stayers on the diagonal) and R_j the source's WRLURI 2006 standardized across the 48 states. It is standardized across the 1,008 state-years. Column (4) builds it from conditional shares instead, $\sum_{j \neq i} \widetilde{\psi}_{ij,t} R_j$ with $\widetilde{\psi}_{ij,t} = \psi_{ij,t}/o_{i,t}$, standardized the same way. Column (3) adds migration openness $o_{i,t} = 1 - \psi_{ii,t}$. Column (2) adds the controls of Table 2, the steep land share, log average distance and lagged log population. Own WRLURI 2006 is the state's standardized index. Entries are coefficients per standard deviation of the regressor, in percentage points of the first-component share in columns (1) to (4) and in loading units in column (5). Column (1) is the specification the diagnostic's decision rule reads. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 Webb draws). Brackets contain 95-percent intervals from inverting that test.

OA.2 Directional Regressions Across States

OA.2.1 The controls entered one at a time

Table OA.3 builds the regressions of Table 2 one control at a time, with every coefficient and standard error, for factor-adjusted log TO (Panel A) and log NET (Panel B) and for raw log FROM (Panel C). Column (5) replaces lagged population, which falls inside each price window and may respond to regulation, by the state's 1980 population from the same BEA series. The 1980 value precedes every endpoint, although it lies inside the windows ending in 1985–1989. The factor-adjusted TO association is 7.97 (2.36) without any population control, 7.50 (2.32) with lagged population and 7.55 (2.35) with 1980 population, and log NET is 6.86, 5.83 and 5.76. For raw log FROM the regulation estimate is 2.97 percent with regulation alone, 2.89 once the steep land share enters, 2.70 with average distance, 2.76 with lagged population, the headline, and 2.77 with 1980 population. Table 2 itself reports every coefficient of the headline specification.

Migration openness exists only from 2005, and entering it would confine the directional regressions to half their span. On the 2005–2025 endpoints, with the state means of log migration openness and lagged log GDP in place of population, the raw-network regulation coefficient is 14.3 percent (standard error 5.8) for log TO, 3.0 (0.6) for log FROM and 11.3 (5.8) for log NET, against 10.7, 3.1 and 7.7 with population on the same endpoints.

Table OA.3: Directional connectedness with the controls entered one at a time, 1985–2025

	Controls entered one at a time				1980 population
	(1)	(2)	(3)	(4)	(5)
Panel A. Factor-adjusted network, log *TO*					
WRLURI 2006	9.59***	9.52***	7.97***	7.50***	7.55***
	(2.11)	(2.11)	(2.36)	(2.32)	(2.35)
Steep land share		0.90	−1.02	−0.71	−0.87
		(2.39)	(2.52)	(2.39)	(2.44)
log average distance			16.92	18.69	19.85
			(13.02)	(12.94)	(13.12)
Lagged log population				3.29	
				(2.29)	
log population in 1980					2.53
					(2.43)
Within R^2	0.060	0.060	0.065	0.072	0.069
Panel B. Factor-adjusted network, log *NET*					
WRLURI 2006	6.98***	7.05***	6.86***	5.83**	5.76**
	(2.18)	(2.20)	(2.38)	(2.08)	(2.12)
Steep land share		−0.95	−1.19	−0.51	−0.79
		(2.34)	(2.77)	(2.28)	(2.39)
log average distance			2.13	6.00	9.70
			(13.44)	(12.06)	(12.71)
Lagged log population				7.18**	
				(2.62)	
log population in 1980					6.52**
					(2.76)
Within R^2	0.026	0.027	0.027	0.053	0.047
Panel C. Raw network, log *FROM*					
WRLURI 2006	2.97***	2.89***	2.70***	2.76***	2.77***
	(0.45)	(0.44)	(0.64)	(0.60)	(0.60)
Steep land share		0.98**	0.75	0.72	0.73
		(0.52)	(0.45)	(0.46)	(0.46)
log average distance			2.03	1.82	1.57
			(3.58)	(3.43)	(3.41)
Lagged log population				−0.38	
				(0.67)	
log population in 1980					−0.40
					(0.69)
Within R^2	0.049	0.054	0.055	0.056	0.056
Year fixed effects	✓	✓	✓	✓	✓
State-years	1,968	1,968	1,968	1,968	1,968

Note: 48 states and 41 annual networks 1985–2025 (1,968 state-years), benchmark network. Panels A and B use the factor-adjusted network and Panel C the raw network. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Column (4) is Table 2, column (4) in Panel A, column (6) in Panel B and column (2) in Panel C. Lagged population is BEA resident population in the calendar year before the endpoint, which lies inside the endpoint's 120-month price window. Column (5) replaces it with log BEA resident population in 1980, from the same series. The year 1980 precedes every endpoint and lies inside the windows of the endpoints 1985 to 1989. $\log \text{NET} = \log(\text{TO}/\text{FROM})$, so its coefficients equal the TO minus the FROM coefficients exactly. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome, and on a logged control it is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

OA.2.2 Functional forms

Table OA.4 reports the regressions of Table 2 with the outcomes in percentage points, and a fractional-response model for external connectedness, in the raw network (Panel A) and the factor-adjusted one (Panel B). In the raw network regulation is associated with 2.12 (0.41) points more external connectedness by least squares and 2.17 (0.45) as a fractional average partial association, against a mean of 78.06, so the log and level estimates agree once scaled. It is associated with 8.0 (2.9) points more TO connectedness and with a net position 5.9 (2.8) points higher, whose interval, unlike that of log NET, just includes zero. In the factor-adjusted network the three are 4.9 (1.6) points of TO, 1.1 (0.6) of external connectedness and 3.8 (1.5) of NET, TO minus FROM. The fractional column is not linear, so it keeps its state-cluster bootstrap standard error, with a t_{47} reference distribution.

Table OA.4: Directional connectedness in levels, raw and factor-adjusted networks, 1985–2025

	TO (1)	FROM (2)	NET (3)	FROM (4)
Panel A. Raw network				
WRLURI 2006	8.011**	2.123***	5.889*	2.169***
	(2.907)	(0.409)	(2.779)	(0.454)
Steep land share	0.419	0.435	−0.016	0.421
	(2.582)	(0.340)	(2.496)	(0.470)
log average distance	13.355	1.416	11.938	1.395
	(14.086)	(2.378)	(13.509)	(2.788)
Lagged log population	8.905**	−0.272	9.177**	−0.283
	(2.827)	(0.477)	(3.017)	(0.529)
Within R^2	0.086	0.067	0.063	–
Pseudo-R^2	–	–	–	0.534
Panel B. Factor-adjusted network				
WRLURI 2006	4.889***	1.067	3.822**	1.087
	(1.613)	(0.649)	(1.523)	(0.703)
Steep land share	−0.547	−0.148	−0.399	−0.147
	(1.505)	(0.524)	(1.433)	(0.645)
log average distance	12.967	8.087**	4.880	8.077**
	(8.817)	(3.280)	(8.254)	(3.808)
Lagged log population	2.303	−2.629***	4.932**	−2.654***
	(1.567)	(0.593)	(1.758)	(0.624)
Within R^2	0.069	0.090	0.054	–
Pseudo-R^2	–	–	–	0.155
Estimator	OLS	OLS	OLS	Fractional logit
Year fixed effects	✓	✓	✓	✓
State-years	1,968	1,968	1,968	1,968

Note: 48 states and 41 annual networks 1985–2025 (1,968 state-years), benchmark network. The regressors and year effects of Table 2 with the outcomes in percentage points, in the raw network (Panel A) and the factor-adjusted network (Panel B). The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Columns (1) to (3) are least squares, and the NET coefficients equal the TO coefficients minus the FROM coefficients exactly. Column (4) is a fractional-logit model of the FROM share, reported as the average partial association in percentage points of FROM. Entries are in percentage points of the outcome, per standard deviation of a standardized regressor and per unit change in a logged control. FROM and TO average 78.06 percentage points in the raw network and 65.58 in the factor-adjusted one, and NET averages zero. In columns (1) to (3), parentheses contain cluster-robust standard errors, clustered by state, and *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from wild cluster bootstrap-t p-values that impose the null (9,999 draws). The fractional-logit model is not linear, so column (4) reports the standard deviation of 999 state-cluster pairs-bootstrap draws, with stars from a t distribution with 47 degrees of freedom.

OA.2.3 The regulation coefficient endpoint by endpoint

Table OA.5 lists the per-endpoint regulation coefficients on external connectedness in the raw form of the three own-lag networks, with their 95-percent intervals. Table OA.6 lists the benchmark coefficients on log TO, log FROM and log NET in the raw and factor-adjusted networks, the estimates behind Figure 7.

Table OA.5: The regulation coefficient endpoint by endpoint, 1985–2025

	Benchmark		Own-lags-free		Eight-own-lag	
Annual network	Estimate	95-percent interval	Estimate	95-percent interval	Estimate	95-percent interval
1985	−5.51	[−12.72, 3.27]	−2.15	[−7.31, 2.92]	–	
1986	−7.45	[−14.33, −0.65]	−0.16	[−5.48, 5.05]	−3.90	[−9.70, 1.66]
1987	−3.33	[−10.06, 2.98]	−1.98	[−7.53, 3.48]	−4.05	[−9.46, 1.17]
1988	−3.06	[−11.43, 4.31]	1.93	[−4.31, 7.67]	−1.95	[−8.30, 4.18]
1989	0.16	[−5.44, 5.70]	8.25	[1.60, 14.63]	−0.60	[−7.11, 5.65]
1990	6.41	[−0.09, 13.30]	6.04	[−0.70, 13.15]	0.61	[−4.08, 5.24]
1991	−0.06	[−5.00, 5.09]	3.60	[−2.54, 10.80]	2.10	[−3.22, 7.53]
1992	3.24	[0.23, 6.65]	8.23	[2.06, 14.97]	3.18	[−2.27, 8.78]
1993	4.38	[0.16, 9.06]	5.96	[−0.89, 12.84]	1.41	[−4.45, 7.01]
1994	2.53	[−1.07, 5.97]	4.75	[−2.07, 11.82]	1.72	[−4.46, 7.59]
1995	2.37	[−1.87, 6.44]	4.26	[−1.83, 11.22]	1.92	[−2.86, 6.43]
1996	3.85	[−0.08, 7.40]	9.49	[3.34, 14.94]	2.58	[−4.32, 8.96]
1997	5.16	[0.67, 9.54]	8.39	[2.04, 13.95]	3.72	[−2.90, 9.88]
1998	2.41	[−3.03, 6.83]	6.73	[−0.57, 13.48]	2.59	[−3.85, 8.76]
1999	4.64	[0.39, 7.96]	8.60	[2.51, 14.30]	4.63	[−1.83, 10.41]
2000	8.50	[3.68, 12.96]	13.18	[8.17, 18.32]	9.37	[3.21, 15.78]
2001	6.00	[2.36, 9.83]	11.84	[6.61, 17.29]	10.16	[4.36, 16.75]
2002	4.36	[−1.51, 9.35]	13.11	[8.48, 18.23]	–	
2003	7.89	[3.39, 12.72]	15.55	[10.53, 21.18]	–	
2004	7.60	[3.43, 12.02]	14.26	[8.13, 21.34]	–	
2005	6.09	[2.71, 9.93]	9.73	[0.90, 19.02]	–	
2006	9.06	[5.07, 13.60]	13.00	[5.08, 21.25]	–	
2007	6.27	[3.02, 9.48]	6.79	[−3.07, 15.88]	–	
2008	3.71	[1.72, 5.98]	4.17	[−1.36, 9.51]	–	
2009	3.48	[1.19, 5.93]	6.60	[2.41, 10.68]	–	
2010	1.90	[−0.05, 4.07]	2.22	[−3.00, 6.77]	0.99	[−4.65, 7.28]
2011	2.78	[0.74, 4.96]	2.28	[−3.10, 6.75]	4.57	[−0.60, 10.40]
2012	2.91	[1.39, 4.48]	1.75	[−3.02, 6.17]	4.11	[−0.87, 9.87]
2013	1.96	[0.44, 3.37]	2.64	[−1.34, 6.73]	0.84	[−2.67, 4.23]
2014	1.85	[−0.03, 3.66]	1.80	[−2.17, 5.15]	5.41	[−0.08, 10.11]
2015	2.59	[−0.78, 5.68]	4.97	[−0.49, 9.58]	5.14	[1.58, 8.71]
2016	2.67	[−0.36, 5.64]	1.07	[−5.12, 7.03]	2.20	[−2.35, 7.00]
2017	3.34	[0.01, 6.63]	3.90	[−2.07, 10.16]	3.68	[−3.12, 10.59]
2018	2.65	[−0.43, 5.62]	5.17	[−0.88, 11.42]	4.77	[−2.44, 11.52]
2019	2.99	[−0.22, 6.21]	5.37	[−0.38, 10.99]	3.18	[−3.78, 9.50]
2020	4.61	[2.50, 6.81]	4.31	[−0.81, 9.36]	5.61	[0.50, 11.02]
2021	2.08	[0.30, 3.76]	2.54	[−1.02, 6.11]	6.50	[2.05, 10.90]
2022	0.95	[−0.96, 2.64]	2.22	[−0.67, 4.97]	1.50	[−2.43, 5.34]
2023	0.50	[−0.81, 1.74]	2.21	[−0.39, 4.67]	1.95	[−1.07, 4.84]
2024	0.63	[−0.49, 1.69]	2.12	[0.25, 3.90]	1.43	[−1.53, 4.11]
2025	1.19	[−0.26, 2.58]	2.02	[−0.75, 4.61]	1.91	[−2.99, 6.70]

Note: Each entry is 100 times the WRLURI 2006 coefficient of a separate 48-state cross-section regression of log FROM on the regressors of Table 2 in one annual network. Brackets contain the 95-percent interval that inverts a wild bootstrap-t test imposing the null (9,999 draws; in a cross-section each state is its own cluster). The benchmark intervals come from the same draws as the stars of the raw log FROM column of Table OA.6, so an interval here excludes zero exactly when that entry carries two or three stars. Figure 7 plots the sending-side counterparts, log TO and log NET. The eight-own-lag network has no variance decomposition in the annual networks of 1985 and 2002–2009.

Table OA.6: The regulation coefficient endpoint by endpoint, raw and factor-adjusted benchmark networks, 1985–2025

	log TO				log FROM				log NET			
	Raw		Factor-adjusted		Raw		Factor-adjusted		Raw		Factor-adjusted	
Annual network, endpoint year	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE
1985	−5.04	(7.69)	−6.82	(5.62)	−5.51	(3.66)	−3.96	(3.80)	0.46	(9.31)	−2.86	(6.14)
1986	3.04	(6.48)	0.94	(4.57)	−7.45**	(3.16)	−5.07	(3.09)	10.49	(7.78)	6.01	(5.04)
1987	7.08	(7.08)	8.61	(5.07)	−3.33	(3.10)	−3.08	(3.23)	10.40	(7.81)	11.69**	(5.55)
1988	13.33*	(7.33)	14.68*	(7.21)	−3.06	(3.66)	0.62	(2.23)	16.39*	(8.59)	14.06*	(7.55)
1989	12.50*	(6.94)	19.10**	(7.62)	0.16	(2.71)	2.41	(3.67)	12.33	(8.01)	16.69**	(7.55)
1990	14.54*	(7.21)	5.31	(5.30)	6.41*	(3.09)	2.72	(3.72)	8.12	(8.28)	2.58	(7.08)
1991	26.60***	(5.96)	10.83*	(5.59)	−0.06	(2.52)	−3.71	(3.22)	26.66***	(7.00)	14.54**	(6.66)
1992	15.86*	(6.95)	−4.69	(8.14)	3.24**	(1.58)	1.29	(2.43)	12.62	(7.06)	−5.98	(9.21)
1993	19.92**	(7.39)	11.53*	(5.53)	4.38**	(2.14)	1.66	(2.56)	15.54	(8.52)	9.87*	(5.56)
1994	19.22***	(6.38)	12.91**	(5.53)	2.53	(1.75)	0.90	(2.77)	16.69**	(7.18)	12.01*	(6.84)
1995	23.36**	(6.27)	7.66	(4.92)	2.37	(2.05)	1.80	(2.21)	20.99**	(6.65)	5.86	(4.88)
1996	8.29	(6.33)	6.78	(4.41)	3.85*	(1.76)	3.34	(2.32)	4.44	(6.92)	3.45	(5.15)
1997	11.03	(6.59)	7.33**	(3.05)	5.16**	(2.13)	5.18**	(2.48)	5.87	(7.59)	2.14	(3.68)
1998	6.91	(6.47)	14.28*	(5.63)	2.41	(2.43)	4.06*	(2.05)	4.51	(6.84)	10.22	(6.31)
1999	0.37	(6.10)	20.55***	(5.50)	4.64**	(1.89)	2.01	(2.91)	−4.27	(6.64)	18.54**	(5.99)
2000	−7.13	(8.34)	21.76***	(6.15)	8.50***	(2.26)	3.13	(3.64)	−15.63	(9.02)	18.63***	(5.99)
2001	9.97**	(3.86)	−0.05	(4.53)	6.00***	(1.77)	0.21	(2.87)	3.97	(4.03)	−0.26	(4.63)
2002	10.68**	(4.72)	−3.96	(4.11)	4.36	(2.38)	−5.92**	(2.68)	6.32	(5.70)	1.96	(5.26)
2003	10.63	(6.70)	−2.66	(5.06)	7.89***	(2.16)	1.70	(3.12)	2.74	(7.22)	−4.36	(6.34)
2004	10.36*	(4.80)	4.12	(5.22)	7.60***	(1.99)	1.49	(3.19)	2.77	(4.97)	2.63	(6.11)
2005	15.87*	(7.71)	21.52***	(5.76)	6.09***	(1.68)	3.64	(2.80)	9.78	(8.48)	17.88***	(6.59)
2006	22.46***	(5.70)	9.01*	(5.09)	9.06***	(1.98)	9.25***	(2.44)	13.40**	(5.48)	−0.24	(4.97)
2007	14.35	(14.18)	16.89***	(5.96)	6.27***	(1.52)	7.74**	(2.97)	8.08	(14.73)	9.14	(6.15)
2008	34.47***	(9.97)	10.48	(6.52)	3.71***	(1.05)	10.15***	(2.83)	30.76***	(9.77)	0.33	(6.31)
2009	28.65**	(11.52)	16.77**	(7.66)	3.48***	(1.16)	8.55***	(2.63)	25.17**	(11.87)	8.22	(7.74)
2010	39.21***	(11.74)	10.57	(7.72)	1.90*	(1.00)	3.14	(2.64)	37.31***	(12.13)	7.43	(8.41)
2011	33.89***	(9.69)	15.54**	(6.77)	2.78***	(1.01)	5.26*	(2.71)	31.11**	(9.95)	10.27	(7.27)
2012	25.20**	(9.68)	7.65	(8.87)	2.91***	(0.77)	4.76*	(2.47)	22.29**	(9.90)	2.89	(10.02)
2013	11.58	(10.51)	4.54	(7.41)	1.96**	(0.72)	3.43*	(1.66)	9.62	(10.82)	1.11	(8.35)
2014	9.82	(11.83)	6.67	(7.90)	1.85*	(0.85)	0.28	(2.55)	7.97	(12.25)	6.39	(9.07)

Continued on the next page

Table OA.6 (continued)

	log TO				log FROM				log NET			
	Raw		Factor-adjusted		Raw		Factor-adjusted		Raw		Factor-adjusted	
Annual network, endpoint year	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE
2015	−4.67	(10.84)	−0.34	(4.33)	2.59	(1.47)	0.27	(2.93)	−7.26	(11.42)	−0.60	(4.76)
2016	−3.35	(9.49)	−1.59	(4.24)	2.67*	(1.44)	−5.62	(3.20)	−6.02	(9.86)	4.03	(5.95)
2017	0.23	(9.64)	5.08	(5.04)	3.34**	(1.64)	−5.80*	(3.15)	−3.10	(9.79)	10.88*	(5.68)
2018	−5.01	(8.81)	8.46	(5.14)	2.65*	(1.48)	−2.46	(2.10)	−7.66	(9.13)	10.91*	(5.88)
2019	−3.10	(6.96)	12.13**	(5.34)	2.99*	(1.48)	1.68	(3.10)	−6.09	(7.42)	10.45	(5.87)
2020	−5.37	(11.91)	8.40	(6.56)	4.61***	(1.12)	7.31**	(3.24)	−9.98	(12.18)	1.09	(7.19)
2021	−8.99	(14.25)	7.50	(5.75)	2.08**	(0.80)	4.64	(3.35)	−11.07	(14.83)	2.86	(7.44)
2022	3.84	(12.73)	2.69	(4.81)	0.95	(0.84)	−1.73	(2.78)	2.88	(13.38)	4.42	(5.85)
2023	5.60	(13.10)	2.69	(6.78)	0.50	(0.60)	0.54	(2.92)	5.10	(13.51)	2.15	(8.45)
2024	5.00	(13.19)	−1.07	(5.88)	0.63	(0.53)	3.43	(2.95)	4.36	(13.59)	−4.50	(6.88)
2025	5.07	(13.10)	−2.71	(5.98)	1.19*	(0.66)	−0.70	(3.01)	3.88	(13.45)	−2.01	(6.13)

Note: Each entry is 100 times the WRLURI 2006 coefficient of a separate 48-state cross-section regression, at the endpoint year of the row's annual network, of the column's outcome on the regressors of Table 2 (WRLURI 2006, the steep land share, log average distance and lagged log population, with an intercept), in the raw and the factor-adjusted benchmark networks. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Standard errors are heteroskedasticity-robust, since in a cross-section each state is its own cluster. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild bootstrap-t p-values that impose the null (9,999 draws). The same draws of each cross-section give these stars, the 95-percent intervals of Figure 7 and the benchmark intervals of Table OA.5, each interval inverting that test, so an interval excludes zero exactly when the entry here carries two or three stars. The log TO and log NET columns are the points of Figure 7, and the raw log FROM column is the benchmark column of Table OA.5.

Table OA.7 repeats the per-endpoint and pooled regressions for the eight annual networks whose windows contain the extreme monthly movements of 1981–1983 (Appendix A), with the benchmark networks re-estimated on winsorized growth rates. The early negative coefficients survive.

Table OA.7: The early endpoints with the 1981–1983 extreme months winsorized

Annual network	Published benchmark	Winsorized input
1985	−5.51	−8.25**
	(3.66)	(2.71)
1986	−7.45**	−5.57**
	(3.16)	(2.68)
1987	−3.33	−5.97**
	(3.10)	(2.95)
1988	−3.06	−2.21
	(3.66)	(3.04)
1989	0.16	−0.49
	(2.71)	(2.52)
1990	6.41*	4.75*
	(3.09)	(2.65)
1991	−0.06	1.94
	(2.52)	(1.88)
1992	3.24**	4.25**
	(1.58)	(1.75)
Pooled 1985–1992, year effects	−1.21	−1.46
	(1.92)	(1.41)

Note: Entries are 100 times the WRLURI 2006 coefficient of the per-endpoint 48-state cross-sections of log FROM on the Table 2 regressors, and of the pooled regression with year effects, on the published benchmark networks and on benchmark networks re-estimated with each state's monthly log growth winsorized at its own 1st and 99th percentiles over 1975–2025. Winsorization clips 14 months per state on average, 32 percent of them in 1981–1983; all eight re-estimated networks are dynamically stable. The eight endpoints are those whose 120-month windows contain 1981–1983. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws). The published-benchmark endpoint stars come from the same draws as those of Table OA.6. The sensitivity would be material if the pooled coefficient changed sign or moved by more than one benchmark standard error, or if any 1985–1988 coefficient changed sign; none does.

OA.2.4 Removing the Bai–Ng number of factors

The factor-adjusted network removes one component, and regional common structure survives it (Appendix C). Table OA.8 re-estimates the state benchmark network after removing, within each 120-month window and inside each validation fold, the number of principal components that the IC_{p2} criterion of Bai and Ng (2002) selects on the window's standardized raw growth, with at most eight. The count is chosen once per window and ranges from two to eight, with a median of four. It is highest in the windows ending in 1985–1992 and lowest in those ending in 1993–2004. After this purge the criterion selects no further factor at 38 of the 41 windows, systemwide connectedness averages 56.7 percent against 65.6 after the one-component purge, and every annual network is stable. The regulation gradient on log TO falls to 1.41 (1.14), with a wild bootstrap p-value of 0.255 and a 95-percent interval of $[-1.34, 3.69]$, and the one on log NET to 1.91 (0.96). With two

components removed at every window the TO gradient is 5.36 (2.06), with an interval of [0.92, 9.89]. The within-year cross-state standard deviation of log TO falls from 0.39 after the one-component purge to 0.25 after the Bai–Ng purge, so the stronger purge also compresses the outcome.

Table OA.8: Regulation and directional connectedness after removing the Bai–Ng number of factors, 1985–2025

	One component			Bai–Ng count			Two components		
Dependent variable:	log TO	log FROM	log NET	log TO	log FROM	log NET	log TO	log FROM	log NET
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
WRLURI 2006	7.50***	1.67	5.83**	1.41	−0.50	1.91*	5.36**	1.28	4.08**
	(2.32)	(1.05)	(2.08)	(1.14)	(0.62)	(0.96)	(2.06)	(0.79)	(1.71)
Steep land share	−0.71	−0.19	−0.51	−0.08	−1.14	1.06	−1.97	−1.40	−0.56
	(2.39)	(0.87)	(2.28)	(1.16)	(1.34)	(0.89)	(1.50)	(0.99)	(1.75)
log average distance	18.69	12.69**	6.00	6.72	6.84	−0.12	17.39	8.94*	8.45
	(12.94)	(5.32)	(12.06)	(6.83)	(5.58)	(5.24)	(11.14)	(5.10)	(10.58)
Lagged log population	3.29	−3.89***	7.18**	2.15	−5.03***	7.18***	3.86**	−4.00***	7.85***
	(2.29)	(0.89)	(2.62)	(1.31)	(0.68)	(1.46)	(1.77)	(0.83)	(2.23)
Components removed		1			2–8			2	
Mean DYCI (percent)		65.6			56.7			61.7	
Year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Endpoints	41	41	41	41	41	41	41	41	41
State-years	1,968	1,968	1,968	1,968	1,968	1,968	1,968	1,968	1,968
Within R^2	0.072	0.083	0.053	0.017	0.082	0.064	0.060	0.058	0.058

Note: 48 states, annual networks 1985–2025. Each network is the state benchmark of Table 2 estimated after principal components of standardized monthly growth are removed within each 120-month window. Each series is regressed on an intercept and the component scores, and the VAR is fitted to the residuals. Inside every validation fold the components are re-extracted from that fold's training months alone. Columns (1)–(3) remove one component and repeat columns (4)–(6) of Table 2. Columns (4)–(6) remove the number of components that the IC_{p2} criterion of Bai and Ng (2002) selects from the window's raw growth, with at most eight. That count is 2 at 4, 3 at 9, 4 at 9, 5 at 8, 6 at 5, 7 at 2 and 8 at 4 endpoints (mean 4.6). Columns (7)–(9) remove two components at every window. The components removed account on average for 45.9, 71.1 and 58.0 percent of the variance of standardized growth in columns (1)–(3), (4)–(6) and (7)–(9). The fitted VAR is stable at every endpoint of the three networks. Mean DYCI is the average over the endpoints of the systemwide share of forecast-error variance attributed to other states. $\log \mathrm{NET} = \log(\mathrm{TO}/\mathrm{FROM})$ is positive for a net source, and its coefficients equal the TO minus the FROM coefficients exactly. Population is BEA state population in the calendar year before the endpoint. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome. On a logged control it is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws). The 95-percent intervals that invert this test for the WRLURI 2006 coefficient on log TO are [2.67, 12.70], [−1.34, 3.69] and [0.92, 9.89] in columns (1), (4) and (7).

OA.2.5 Amplitude and tracking decomposition

Table OA.9 adds two variables to the regressions of Table 2, in the raw and the factor-adjusted networks, each measured on the same 120-month window as the network. They are the share of a state's growth variance taken by the first principal component of the standardized 48-series panel, and the log standard deviation of its VAR innovation. Both are built from the price series whose forecast-error variance decomposition defines the outcome, and both are plausibly consequences of regulation, so the table is a decomposition of the regulation associations and not a specification of

them. It asks how much of each association runs through the size and the national comovement of a state's price surprises.

In the raw network both variables matter. A 1 percent larger innovation standard deviation is associated with about 0.74 percent more TO connectedness and 0.17 percent less external connectedness, and a state whose growth loads more heavily on the first principal component has more external connectedness, 0.27 percent per percentage point of explained variance. Holding both fixed cuts the external-connectedness association from 2.76 to 1.12 percent (0.68), the TO association from 10.5 to 5.3 (3.5) and log NET from 7.8 to 4.2 (3.4), none of them then distinguishable from zero. In the factor-adjusted network the TO association is 7.5 percent (2.3) without them and 6.8 (2.3) with them, so the sending side beyond the national cycle does not run through the size of a state's own surprises. Factor-adjusted log NET falls from 5.8 (2.1) to 3.1 (1.8). The factor-adjusted external-connectedness association rises from 1.7 (1.0) to 3.6 (1.1) once the two variables enter. That rise should not be read as a receiving-side result, because in the factor-adjusted network the first-component share measures how much of each state's variation the purge removed, so conditioning on it conditions on the purge itself.

Table OA.9: Directional connectedness with the amplitude controls added, raw and factor-adjusted networks, 1985–2025

	Raw			Factor-adjusted		
	log TO	log FROM	log NET	log TO	log FROM	log NET
	(1)	(2)	(3)	(4)	(5)	(6)
WRLURI 2006	5.35	1.12	4.23	6.77^{***}	3.65^{***}	3.13^{*}
	(3.54)	(0.68)	(3.38)	(2.30)	(1.12)	(1.79)
First-component variance share (pp)	0.36^{***}	0.27^{***}	0.09	−0.02	-0.18^{***}	0.17^{**}
	(0.11)	(0.02)	(0.11)	(0.08)	(0.03)	(0.07)
log innovation SD	73.56^{***}	-17.12^{***}	90.68^{***}	48.58^{***}	-27.81^{***}	76.38^{***}
	(6.55)	(1.43)	(7.01)	(4.82)	(2.29)	(5.36)
Steep land share	0.43	0.68	−0.25	−1.28	0.25	−1.53
	(2.96)	(0.54)	(2.79)	(2.72)	(0.96)	(2.86)
log average distance	−5.34	8.33^{**}	−13.67	−0.15	23.04^{***}	−23.19
	(14.95)	(3.24)	(14.06)	(12.94)	(5.74)	(12.63)
Lagged log population	−4.84	2.74^{***}	-7.59^{*}	-6.72^{**}	2.29^{**}	-9.00^{***}
	(3.87)	(0.69)	(3.92)	(2.05)	(0.91)	(1.93)
Year fixed effects	✓	✓	✓	✓	✓	✓
Within R^2	0.318	0.413	0.358	0.243	0.336	0.398

Note: 48 states and 41 annual networks 1985–2025. Columns (1)–(3) use the raw benchmark network and columns (4)–(6) the factor-adjusted benchmark network. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Columns report the regressions of Table 2 with two amplitude controls added, each measured on the network's own 120-month window. The first-component variance share is the state's share of growth variance on the first principal component of the standardized 48-series panel, in percentage points; it is computed from raw growth and so is the same in both networks. The log innovation standard deviation (SD) is that of the state's VAR innovation in the network. Both are functions of the price series that define the outcome, so this is a decomposition of the headline associations rather than a specification of them. In the factor-adjusted network the first-component share measures what the purge removed, so conditioning on it conditions on the purge itself. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome; on a logged control, it is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

OA.3 The Labor-Demand Interaction

Table 3 of the paper reports the interaction of regulation and the labor-demand shock on the metropolitan receiving side, log FROM. Table OA.10 reports the same four columns on the sending side, log TO, with the same 100 areas, controls and shock, standardized across areas within each window. At the average shock the regulation gradient of TO is 2.9 (1.7), and it rises with the shock by 1.9 (0.9), with a wild bootstrap p-value of 0.05 (column 1). Beside regulation the steep land share interaction is 0.6 (1.1) (column 3). Split at the average shock, the regulation interaction is 2.6 (1.7) above the average and 0.8 (1.9) below it (column 4), and neither their difference, 1.8 (2.9), nor their sum, 3.4 (2.1), is distinguishable from zero.

Table OA.11 sets the checks of column (1) side by side for the two sides of an area's position. On the receiving side every check of column (1) leaves the interaction between 0.63 and 0.96, with p-values below 0.02, and it holds within areas over time. On the sending side the estimates stay between 1.4 and 2.5 when areas or endpoints are left out, but ten of those thirteen checks have p-values above 0.05, and within areas over time the interaction is 0.8 (1.1). Neither side shows the interaction in the raw network, and across states both are imprecise.

Table OA.12 repeats the comparison of Table 8 on an apportioned node set of 75 areas, built by the same rule with 75 seats. It gives each of the 48 states at least one area and no state more than four. The receiving-side interaction is 0.82 (0.42), and the slope above the average shock 1.39 (0.58). The sending-side interaction is 0.59 (1.24). Among pairs in different states the factor-adjusted allocation coefficient of column (4) is 0.062 (0.022).

OA.3.1 The sending-side test with an era-scaled shock

The tables of this subsection measure the labor-demand shock relative to the year's average in units of the within-year standard deviation averaged over each industry-classification era (Appendix F.1), instead of standardizing it within each window, and report the sending-side test with that shock. Table OA.13 splits the metropolitan interaction of regulation and the labor-demand shock at the year's average shock, together with the split at the within-year median and the same regressions in the raw network. Every column holds fixed the number of an area's same-state sample partners. On total TO the median split gives a difference of 1.8 (2.6) and a sum of 4.0 (1.8), against 1.7 and 4.0 at the average. Without the partner control the average split on total TO gives 6.7 above and -1.2 below, a difference of 7.9 (3.2), as the note reports. In the raw network the interaction is -0.8 (1.1) on total TO and -1.0 (1.3) on TO to other states, and no difference or sum of the split slopes is distinguishable from zero.

Table OA.14 reports two further checks. Its first panel repeats the regressions with the outcomes in percentage points, where the parts of TO going to other states' areas and to same-state areas add up to the total exactly. The interaction on the total, 1.11 points, is the sum of 1.57 in the other-state part and -0.46 in the same-state part, and the difference between the split slopes, 1.99, is the sum of 2.93 and -0.94, so the other-state part carries more than the whole. None of these

Table OA.10: Regulation, terrain and relative labor demand on the metropolitan sending side, factor-adjusted network, 1985–2025

Dependent variable	log TO			
	(1)	(2)	(3)	(4)
WRLURI 2006	2.92		3.11	2.38
	(1.69)		(1.77)	(2.16)
WRLURI × S	1.86*		1.65	
	(0.85)		(1.00)	
WRLURI × S^+				2.59
				(1.72)
WRLURI × S^-				0.81
				(1.85)
Steep land share		0.76	1.14	1.50
		(2.09)	(1.95)	(2.23)
Steep land share × S		0.81	0.58	
		(0.97)	(1.07)	
Steep land share × S^+				0.29
				(1.14)
Steep land share × S^-				1.24
				(2.36)
Labor-demand shock S	−2.33**	−2.61**	−2.39**	
	(1.08)	(1.09)	(1.06)	
S^+				−3.38**
				(1.61)
S^-				−1.53
				(1.93)
Controls and same-state sample partners	✓	✓	✓	✓
Climate amenity index	✓	✓	✓	✓
Year and Census-division effects	✓	✓	✓	✓
Areas	100	100	100	100
Area-years	4,100	4,100	4,100	4,100
Within R^2	0.288	0.283	0.288	0.289

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025, factor-adjusted benchmark network, estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The dependent variable is log TO, the log of the sum, over the other areas, of the shares of their forecast-error variance attributed to an area. The columns are those of Table 3, which reports log FROM. The labor-demand shock S is predicted labor demand, the shift-share prediction of an area's earnings growth over the window (Appendix F.1), standardized across areas in each annual network, so that it has mean zero and standard deviation one in every window. $S^+ = \max(S, 0)$ and $S^- = \min(S, 0)$. WRLURI 2006 and the steep land share, the share of the area's land with a slope above 15 percent (Lutz and Sand, 2026), are standardized across areas. Controls are the climate amenity index, lagged log population, the outside-node-set share and the number of an area's same-state sample partners, with year and Census-division effects. The climate amenity index is the sum of the four climate components of the USDA natural amenity scale (McGranahan, 1999), January temperature, January sun, July temperature and July humidity, without its topography and water-area components, averaged over an area's counties with 1990 population weights and standardized across areas. In column (4) the WRLURI slope above the average minus that below it is 1.78 (2.87, $p = 0.568$) and their sum is 3.39 (2.14, $p = 0.182$). For the steep land share they are −0.95 (2.80, $p = 0.776$) and 1.53 (2.44, $p = 0.717$). Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in TO. Within R^2 is computed against the outcome demeaned by whatever the design absorbs, because the unadjusted R^2 of these models is mostly the fixed effects. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws). The regressions run in Stata (reghdfe, boottest).

Table OA.11: Checks of the labor-demand interaction on the receiving and the sending side, metropolitan areas and states, 1985–2025

	log FROM		log TO		
	WRLURI × S (SE)	p	WRLURI × S (SE)	p	Clusters / obs.
Panel A. Inference and specification					
Table 3, column (1)	0.76^{***} (0.21)	0.001	1.86^{*} (0.85)	0.050	100 / 4,100
Clusters by dominant industry in 1975	0.76^{***} (0.16)	0.010	1.86^{*} (0.69)	0.055	16 / 4,100
Column (4), WRLURI × S^{+}, industry clusters	1.24^{**} (0.55)	0.037	2.59 (1.95)	0.171	16 / 4,100
Full natural-amenity scale and its interaction added	0.85^{***} (0.24)	0.003	2.11^{*} (1.02)	0.068	100 / 4,100
Panel B. Areas left out					
Without East North Central	0.70^{***} (0.23)	0.007	1.87^{*} (0.89)	0.065	87 / 3,567
Without East South Central	0.71^{***} (0.22)	0.003	1.77^{**} (0.80)	0.044	93 / 3,813
Without Middle Atlantic	0.74^{***} (0.22)	0.003	1.47 (0.88)	0.117	88 / 3,608
Without Mountain	0.63^{***} (0.21)	0.005	2.15^{**} (0.83)	0.029	92 / 3,772
Without New England	0.86^{***} (0.22)	0.000	2.31^{**} (0.89)	0.027	92 / 3,772
Without Pacific	0.83^{***} (0.23)	0.002	1.80 (1.04)	0.106	86 / 3,526
Without South Atlantic	0.76^{***} (0.23)	0.003	1.38 (1.06)	0.249	79 / 3,239
Without West North Central	0.94^{***} (0.21)	0.001	1.78^{*} (0.93)	0.077	94 / 3,854
Without West South Central	0.67^{***} (0.24)	0.005	1.91 (1.02)	0.114	89 / 3,649
Without San Jose, San Francisco, Seattle, Boston, Austin	0.68^{***} (0.22)	0.005	1.56 (0.95)	0.142	95 / 3,895
Panel C. Endpoints					
SIC endpoints	0.96^{***} (0.27)	0.003	1.89 (1.65)	0.317	100 / 1,600
NAICS endpoints	0.74^{**} (0.23)	0.013	2.46 (1.31)	0.136	100 / 1,500
Endpoints outside the chained windows	0.86^{***} (0.17)	0.000	1.64^{*} (0.86)	0.072	100 / 3,100
Panel D. Other margins					
Area effects, factor-adjusted	0.69^{*} (0.28)	0.055	0.84 (1.09)	0.472	100 / 4,100
Area effects, raw	0.04 (0.21)	0.853	−1.28 (1.22)	0.289	100 / 4,100
Raw network	0.01 (0.16)	0.955	−0.87 (1.09)	0.428	100 / 4,100
48 states, factor-adjusted	1.35 (1.04)	0.247	−1.50 (2.36)	0.560	48 / 1,968
48 states, raw	0.86 (0.69)	0.242	−2.07 (3.34)	0.563	48 / 1,968

Note: Each row is the interaction of WRLURI 2006 with the labor-demand shock S in column (1) of Table 3 (log FROM) and of Online Appendix Table OA.10 (log TO), re-estimated as the row describes, unless the row names column (4). Rows without a network are factor-adjusted. Clusters by dominant industry group the areas by the industry with the largest wage-bill share in 1975, 16 clusters, with year and Census-division effects entered as dummies. The amenity row adds the full natural-amenity scale of McGranahan (1999) and its interaction with the shock to the climate control every column carries. Endpoint rows keep the shock of the full sample; the chained windows are those that span the 1997 change of industry classification. Area effects replace the Census-division effects. The state rows use the controls of Table 2 with year effects and clusters by state. Entries are 100 times the coefficients, with cluster-robust standard errors in parentheses; p is the wild cluster bootstrap-t p-value that imposes the null (9,999 draws), and stars follow it: *, ** and *** denote $p < 0.10$, 0.05 and 0.01. The last column gives clusters and observations. The regressions run in Stata (reghdfe, boottest).

Table OA.12: Metropolitan results on the benchmark and an apportioned node set of 75 areas

	Benchmark, 100 largest		Apportioned, 75	
	Raw (1)	Factor-adjusted (2)	Raw (3)	Factor-adjusted (4)
Panel A. Table 3, log *FROM*				
WRLURI × S, column (1)	0.01	0.76***	−0.03	0.82*
	(0.16)	(0.21)	(0.33)	(0.42)
WRLURI × S^+, column (4)	0.54**	1.24**	0.65	1.39*
	(0.26)	(0.37)	(0.47)	(0.58)
WRLURI × S^-, column (4)	−0.63	0.19	−0.58	0.25
	(0.30)	(0.41)	(0.58)	(0.90)
Panel B. The same regressions, log *TO*				
WRLURI × S, column (1)	−0.87	1.86*	−2.09	0.59
	(1.09)	(0.85)	(1.37)	(1.24)
WRLURI × S^+, column (4)	−2.42	2.59	−3.03	1.44
	(2.31)	(1.72)	(2.79)	(1.81)
WRLURI × S^-, column (4)	0.39	0.81	−1.39	0.28
	(2.58)	(1.85)	(2.91)	(2.59)
Areas, Panels A and B	100	100	74	74
Panel C. Table 4, column (4), migration exposure				
All pairs	0.060***	0.171***	0.056***	0.186***
	(0.022)	(0.056)	(0.019)	(0.053)
Pairs in different states	0.009	0.060**	0.020**	0.062***
	(0.010)	(0.025)	(0.009)	(0.022)
Pairs in different states, area-year pairs	200,256	200,256	115,038	115,038
States with an area	39	39	48	48
Most areas in one state	11	11	4	4

Note: Columns (1)–(2) are the benchmark node set, the 100 largest contiguous-US metropolitan areas by 2000 population. Columns (3)–(4) re-estimate both networks, with the benchmark specification, on an apportioned node set of 75 areas that gives each of the 48 contiguous states at least one area and divides the rest by the Huntington–Hill method on the square root of each state's metropolitan population, taking the largest areas first within a state. It shares 64 areas with the benchmark. Every second-stage variable is rebuilt on the node set's own network, and the supply proxies and the labor-demand shock are standardized within it. Panel A repeats the regressions of Table 3, with the same controls, including the number of same-state sample partners, on the areas with a metropolitan Wharton index value, which the survey does not provide for Cheyenne, Wyoming, an area of the apportioned node set, and Panel B the same regressions with log TO (Table OA.10). Their entries are 100 times the coefficients, with standard errors clustered by area and wild cluster bootstrap-t p-values that impose the null (9,999 draws). Panel C repeats column (4) of Table 4, 2005–2025, with its interacted geography controls and with standard errors that allow dependence between pairs sharing an area and p-values from $t(G-1)$, G the number of areas. *, ** and *** denote $p < 0.10$, 0.05 and 0.01. Panels A and B and the apportioned columns of Panel C run in Stata, whose version of the Panel C estimator reproduces the benchmark cells.

Table OA.13: The metropolitan interaction split at the year's average labor-demand shock, in full, at the median and on the raw network, 1985–2025

Dependent variable:	log TO			
	All 100 areas		82 areas with a partner	
	Total	To other states' areas	To same-state areas	Total
	(1)	(2)	(3)	(4)
Panel A. Split at the year's average, factor-adjusted network				
WRLURI 2006	2.04	1.61	−0.75	0.57
	(1.99)	(1.98)	(2.86)	(1.66)
Shock above the average, S^+	−2.65	−0.79	−4.02	−0.91
	(1.68)	(2.37)	(2.90)	(1.43)
Shock below the average, S^-	−1.59	−2.04	1.97	−0.76
	(1.91)	(2.36)	(2.22)	(1.65)
WRLURI × S^+, b^+	2.86	4.76*	−2.23	3.15*
	(1.55)	(2.06)	(2.73)	(1.20)
WRLURI × S^-, b^-	1.17	0.89	1.06	−1.13
	(1.60)	(1.99)	(2.65)	(1.57)
Difference, $b^+ - b^-$	1.70	3.87	−3.29	4.28
	(2.54)	(3.25)	(4.25)	(2.26)
Sum, $b^+ + b^-$	4.03*	5.64**	−1.16	2.03
	(1.86)	(2.43)	(3.30)	(1.64)
Panel B. Split at the within-year median, factor-adjusted network				
WRLURI × shock above the median	2.93*	4.93*	−2.44	3.23*
	(1.56)	(2.09)	(2.79)	(1.19)
WRLURI × shock below the median	1.12	0.85	0.89	−1.13
	(1.59)	(1.98)	(2.61)	(1.56)
Difference, above − below	1.81	4.08	−3.33	4.36
	(2.56)	(3.27)	(4.29)	(2.26)
Sum, above + below	4.05**	5.78**	−1.54	2.11
	(1.85)	(2.42)	(3.30)	(1.62)
Panel C. Raw network				
WRLURI × shock	−0.80	−1.04	−1.12	−0.88
	(1.14)	(1.27)	(1.52)	(1.36)
WRLURI × S^+, b^+	−1.88	−2.11	−2.64	−0.66
	(2.32)	(2.46)	(2.16)	(2.34)
WRLURI × S^-, b^-	0.30	0.10	0.79	−1.33
	(2.14)	(2.44)	(2.92)	(2.51)
Difference, $b^+ - b^-$	−2.18	−2.20	−3.43	0.67
	(3.78)	(4.12)	(4.14)	(3.95)
Sum, $b^+ + b^-$	−1.58	−2.01	−1.85	−1.99
	(2.37)	(2.65)	(3.03)	(2.82)
Areas	100	100	82	82
Area-years	4,100	4,100	3,362	3,362

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025. Each column is the regression of Table 3 with the same column number, with its controls and its year and Census-division effects. The factor-adjusted benchmark network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The labor-demand shock S is split at the year's average into $S^+ = \max(S, 0)$ and $S^- = \min(S, 0)$, each with its own main effect and its own interaction with WRLURI, b^+ and b^-. The WRLURI row is the regulation gradient at the year's average shock. The difference and sum rows come from the same regression reparametrized so that each contrast is one coefficient. Panel B places the split at the within-year median. Panel C repeats the regressions on the raw benchmark network, with TO and its two parts built in the same way. Without the same-state-partner control, column (1) gives b^+ of 6.73 (2.02), b^- of −1.15 (1.99), a difference of 7.88 (3.24) and a sum of 5.57 (2.35), and the Panel A interaction of Table 3 is 3.02 (1.13). Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

is distinguishable from zero at the 5-percent level in points. Its second panel drops the endpoints 2001–2010, whose windows chain SIC growth up to 2000 to NAICS growth from 2001. On the 31 other endpoints the interaction is 1.77 (0.87) on total TO and 2.40 (1.12) on TO to other states, with p-values of 0.07.

Table OA.14: The metropolitan interaction in percentage points and outside the chained windows, factor-adjusted network, 1985–2025

Dependent variable:	TO in percentage points (Panel A) and log TO (Panel B)			
	All 100 areas		82 areas with a partner	
	Total	To other states' areas	To same-state areas	Total
	(1)	(2)	(3)	(4)
Panel A. TO in percentage points, all 100 areas at 41 endpoints				
WRLURI × shock	1.11	1.57*	−0.46	–
	(0.63)	(0.70)	(0.35)	
WRLURI × S^+, b^+	2.11*	2.95*	−0.84	–
	(1.08)	(1.32)	(0.67)	
WRLURI × S^-, b^-	0.12	0.02	0.10	–
	(1.23)	(1.17)	(0.42)	
Difference, $b^+ - b^-$	1.99	2.93	−0.94	–
	(1.88)	(2.03)	(0.93)	
Sum, $b^+ + b^-$	2.23	2.97*	−0.74	–
	(1.34)	(1.45)	(0.62)	
Areas	100	100	100	–
Area-years	4,100	4,100	4,100	–
Panel B. 31 endpoints outside the chained windows				
WRLURI × shock	1.77*	2.40*	−1.09	0.88
	(0.87)	(1.12)	(1.64)	(0.80)
WRLURI × S^+, b^+	3.47**	4.71*	−1.47	3.28**
	(1.64)	(2.16)	(2.52)	(1.25)
WRLURI × S^-, b^-	0.11	0.03	−0.57	−1.92
	(1.65)	(1.94)	(2.49)	(1.66)
Difference, $b^+ - b^-$	3.36	4.68	−0.90	5.20*
	(2.64)	(3.33)	(3.88)	(2.43)
Sum, $b^+ + b^-$	3.58	4.74*	−2.04	1.36
	(1.96)	(2.40)	(3.17)	(1.66)
Areas	100	100	82	82
Area-years	3,100	3,100	2,542	2,542

Note: The 100 largest metropolitan areas, annual networks 1985–2025. Each column is the regression of Table 3 with the same column number, with its controls and its year and Census-division effects, on the factor-adjusted network. The factor-adjusted benchmark network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The labor-demand shock S is split at the year's average into $S^+ = \max(S, 0)$ and $S^- = \min(S, 0)$, with WRLURI interactions b^+ and b^-, as in Table 3. In Panel A the outcomes are TO and its two parts in percentage points on all 100 areas, with TO to same-state areas equal to zero for the 18 areas without a same-state partner. Each coefficient in column (1) therefore equals the sum of those in columns (2) and (3), and column (4) has no counterpart. Panel B leaves out the ten endpoints 2001–2010, whose shock windows cross the SIC–NAICS change in the BEA county data. Their shocks are chained, SIC growth up to 2000 and NAICS growth from 2001, and the 2000–2001 year is omitted. The shock of the window ending 2001 is SIC growth 1991–2000. In Panel A an entry is the coefficient, in percentage points of TO per standard deviation. In Panel B entries are 100 times the coefficients, approximately the percent difference in TO per standard deviation. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

Table OA.15 repeats the inference with clusters that allow dependence among areas with similar industry mix. Each area is assigned its dominant industry, the largest wage-bill share among the shift-share cells in 1975, the base year of the first window, which gives 16 clusters, and alternatively in 2001, the first NAICS base year, which gives 12. With the 1975 clusters the p-value of the interaction rises from 0.043 to 0.074 on total TO and to 0.222 on TO to other states, and that of the sum of the split slopes on TO to other states from 0.049 to 0.218. With the 2001 clusters the p-values are 0.061 and 0.027. The precision of the metropolitan interaction therefore depends on how dependence among areas with similar industry mix is treated.

Table OA.16 describes the shock itself. Predicted ten-year wage-bill growth is positive in every area-window in nominal terms, with a mean of 47.6 log points. Deflated by the CPI-U over the same ten years it averages 17.1 and is negative in 24 of the 4,100 area-windows, 0.6 percent, from 9 areas, and in 1.2 percent of the below-average ones. The split at the year's average therefore compares above- with below-average growth, not rising with falling demand. Panel B reports the regulation gradient of TO by within-year tercile of the shock, with standard errors. With the partner control the middle tercile has the steepest gradient, and without it the top tercile, and the difference between the top and bottom terciles is not distinguishable from zero in any column.

Table OA.15: Inference for the metropolitan interaction clustered by area and by dominant industry, factor-adjusted network, 1985–2025

Dependent variable:	log TO			
	All 100 areas		82 areas with a partner	
	Total	To other states' areas	To same-state areas	Total
	(1)	(2)	(3)	(4)
Panel A. WRLURI × shock				
Estimate	1.97	2.90	−0.96	1.24
Standard error, clustered by area	(0.86)	(1.17)	(1.80)	(0.87)
by dominant 1975 SIC industry	(0.73)	(1.32)	(1.37)	(1.02)
by dominant 2001 NAICS industry	(0.77)	(1.10)	(1.74)	(0.40)
p-value, clustered by area	0.043	0.043	0.663	0.272
by dominant 1975 SIC industry	0.074	0.222	0.598	0.767
by dominant 2001 NAICS industry	0.061	0.027	0.652	0.183
95-percent interval, 1975 SIC industry	[−0.36, 4.64]	[−2.13, 6.34]	[−4.95, 3.85]	[−2.03, 4.98]
2001 NAICS industry	[−0.54, 5.23]	[0.97, 7.99]	[−19.22, 1.62]	[−2.40, 2.29]
Panel B. Difference, $b^+ - b^-$				
Estimate	1.70	3.87	−3.29	4.28
Standard error, clustered by area	(2.54)	(3.25)	(4.25)	(2.26)
by dominant 1975 SIC industry	(2.92)	(3.47)	(2.75)	(1.88)
by dominant 2001 NAICS industry	(2.96)	(2.71)	(3.87)	(2.22)
p-value, clustered by area	0.521	0.262	0.533	0.102
by dominant 1975 SIC industry	0.632	0.357	0.442	0.152
by dominant 2001 NAICS industry	0.610	0.249	0.553	0.283
95-percent interval, 1975 SIC industry	[−8.58, 14.52]	[−11.38, 17.90]	[−12.67, 4.70]	[−4.47, 10.90]
2001 NAICS industry	[−11.66, 7.57]	[−7.12, 12.70]	[−31.94, 5.51]	[−8.31, 8.81]
Panel C. Sum, $b^+ + b^-$				
Estimate	4.03	5.64	−1.16	2.03
Standard error, clustered by area	(1.86)	(2.43)	(3.30)	(1.64)
by dominant 1975 SIC industry	(1.68)	(2.55)	(2.97)	(1.68)
by dominant 2001 NAICS industry	(1.96)	(2.55)	(2.86)	(1.28)
p-value, clustered by area	0.054	0.049	0.758	0.273
by dominant 1975 SIC industry	0.044	0.218	0.733	0.574
by dominant 2001 NAICS industry	0.113	0.038	0.725	0.352
95-percent interval, 1975 SIC industry	[0.17, 11.25]	[−4.97, 13.83]	[−9.79, 9.35]	[−4.87, 8.83]
2001 NAICS industry	[−1.22, 13.29]	[0.38, 18.71]	[−37.35, 2.62]	[−12.46, 5.79]
Clusters, areas	100	100	82	82
Clusters, dominant 1975 SIC industries	16	16	15	15
Clusters, dominant 2001 NAICS industries	12	12	10	10

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025. Each column is the regression of Table 3 with the same column number, on the factor-adjusted network, and the estimates are those of that table. The factor-adjusted benchmark network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The rows compare cluster-robust standard errors and wild cluster bootstrap-t p-values that impose the null (9,999 Webb draws) clustered by metropolitan area, as in Table 3, with the same statistics clustered by each area's dominant industry. The dominant industry is the industry with the largest earnings share among the shift-share shares of one base year, fixed across annual networks so that each area stays in one cluster. The 1975 SIC shares are those of the first window and give 16 industries, the largest with 42 of the 100 areas. The 2001 NAICS shares are those of the first NAICS base year of the construction and give 12 industries, the largest with 41 areas. The intervals invert the dominant-industry bootstrap tests, whose draws use one seed in every cell. Table 3 gives the area-clustered intervals. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome.

Table OA.16: Predicted labor-demand growth in nominal and real terms, and the metropolitan regulation gradient by shock tercile, 1985–2025

	(1)	(2)	(3)
Panel A. Predicted labor-demand growth B over the window (log points)			
	Nominal	Real	Real (one-interval deflator)
Mean	47.6	17.1	16.5
Standard deviation	15.4	6.7	6.5
10th percentile	29.7	8.1	7.6
Median	46.1	17.5	17.3
90th percentile	69.3	25.6	24.1
Share of area-windows with $B < 0$ (percent)	0.0	0.6	0.8
among below-average area-windows (percent)	0.0	1.2	1.7
Areas with at least one window with $B < 0$	0	9	14
Endpoints with at least one such window	0	8	8
Panel B. Regulation gradient of factor-adjusted log *TO by within-year tercile of the shock*			
	Total	Total	To other states' areas
Same-state sample partners held fixed	–	✓	✓
WRLURI in the bottom tercile	2.36	0.91	0.99
	(2.86)	(2.04)	(2.22)
WRLURI in the middle tercile	5.53	5.11**	5.33**
	(3.54)	(2.18)	(2.25)
WRLURI in the top tercile	6.61	3.03	3.43
	(4.00)	(2.43)	(2.66)
Top minus bottom tercile	4.25	2.12	2.45
	(3.54)	(2.79)	(3.14)
Area-years	4,100	4,100	4,100

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025, the sample of Table 3. Predicted labor demand B is the shift-share prediction of an area's earnings growth over the window of each annual network (Appendix F.1), a change in the log of nominal earnings. Entries in Panel A are 100 times the log change. Real growth subtracts the change in the log of the calendar-year average CPI-U (BLS series CPIAUCSL) over the years whose earnings growth B contains. For the windows ending 2002–2010 these are the years up to 2000 and from 2001, since the chained shock omits the 2000–2001 change, and for the window ending 2001 they are 1991–2000. The one-interval deflator uses the change from ten years before the endpoint to the endpoint (2024 for 2025) for every window. Below-average area-windows are those with a negative labor-demand shock S, predicted labor demand below the year's average across areas. Panel B replaces the shock in Table 3 by indicators of within-year terciles of S, each with its own interaction with WRLURI, so each row is the regulation gradient within one tercile. Column (1) does not hold the number of same-state sample partners fixed, and columns (2) and (3) do, as every column of Table 3 does. The factor-adjusted benchmark network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). In Panel B entries are 100 times the coefficients. Per standard deviation of WRLURI, an entry is approximately the percent difference in TO. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

OA.4 The Metropolitan Receiving Side and the Destination Split

OA.4.1 External connectedness across fixed-effect structures

Table OA.17 extends Table D.3 to year effects alone, to coastal, sunbelt and interior belt effects, and to principal-state effects on all 100 areas. Without the two population controls the regulation association is 1.26 percent with belt effects and 1.31 with division effects. With them it is 0.81 with year effects alone, 0.70 with belt effects, 0.88 with division effects, 0.45 with principal-state effects on all 100 areas and 0.52 on the 64 in states with at least three sample areas. With the controls it is distinguishable from zero at the 5-percent level with division effects and within the 64 areas, and not with year effects alone ($p = 0.05$), belt effects (0.12) or principal-state effects on all 100 areas (0.06). The outside-node-set share is negative and precisely estimated in every column, and log population is unrelated to external connectedness once the outside-node-set share is held fixed. With principal-state-by-year effects in place of principal-state effects the association is 0.51 (0.15) on the 64 areas (Table D.3, column 4) and 0.45 (0.23) on all 100. A wild cluster bootstrap over principal states, which allows arbitrary dependence among the areas of a state, gives p-values of 0.009 for both within-state columns on the 64 areas and of 0.089 and 0.094 on all 100, where states with a single sample area contribute nothing to the estimate.

Table OA.17: Metropolitan external connectedness across fixed-effect structures, 1985–2025

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
WRLURI 2006	0.81*	1.26***	0.70	1.31***	0.88**	0.45*	0.52***
	(0.37)	(0.39)	(0.40)	(0.30)	(0.31)	(0.19)	(0.14)
Steep land share	0.12	0.25	0.16	0.24	0.18	−0.30	−0.23
	(0.24)	(0.24)	(0.23)	(0.47)	(0.43)	(0.16)	(0.16)
Lagged log population	0.05		0.05		−0.05	−0.08	0.07
	(0.27)		(0.26)		(0.26)	(0.15)	(0.13)
Outside-node-set share	−1.63***		−1.42***		−1.60***	−0.84***	−0.71***
	(0.38)		(0.42)		(0.31)	(0.24)	(0.23)
Year fixed effects	✓	✓	✓	✓	✓	✓	✓
Fixed effects beyond year	None	Belt	Belt	Division	Division	Principal state	Principal state
Metropolitan areas	100	100	100	100	100	100	64
Within R^2	0.139	0.047	0.086	0.033	0.075	0.010	0.047

Note: The 100 largest metropolitan areas and 41 annual networks 1985–2025, benchmark network. The dependent variable is the log of the area's external connectedness. Columns (4), (5) and (7) are columns (1), (2) and (3) of Table D.3, Panel A. Belts are coastal, sunbelt and interior. An area's principal state is the state holding the largest share of its 2010 housing units. Column (7) is restricted to the 64 areas in the 12 principal states with at least three sample areas. WRLURI 2006 and the steep land share are standardized within the node set. Population is the area's county population in the calendar year before the endpoint, and the outside-node-set share is the standardized share of the population within 200 km of the area's internal point that lives in counties belonging to no sample area. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in external connectedness, and on lagged log population it is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

OA.4.2 Samples, functional forms and the outside-node-set radius

The regression without the two population controls is Table D.3, column (1). Entering them lowers the regulation association under division effects from 1.31 to 0.88 percent. Table OA.18 varies the rest of the design for the division and within-state columns. The association is 0.89 percent over the 2005–2025 endpoints and 0.88 over 1985–2004 with division effects, and 0.64 and 0.39 within states, the earlier estimates less precise. In percentage points it is 0.77 (0.26) with division effects and 0.48 (0.12) within states, and the fractional-logit average partial associations are 0.79 and 0.46. The one arbitrary element of the outside-node-set control is its 200-kilometer radius. Rebuilt at 100 and 300 kilometers, the regulation association is 1.11 and 0.97 percent with division effects and 0.55 and 0.48 within states, distinguishable from zero in every case. The control itself is negative at every radius and weakest at 100 kilometers, where much of what lies around an area's internal point is the area's own counties.

Table OA.18: The metropolitan regulation coefficient across samples, functional forms and radii

	Division effects (1)	Within states (2)
Base design, log outcome, 1985–2025	0.88**	0.52***
	(0.31)	(0.14)
2005–2025 endpoints	0.89***	0.64***
	(0.23)	(0.18)
1985–2004 endpoints	0.88	0.39
	(0.54)	(0.19)
Levels, percentage points	0.77**	0.48***
	(0.26)	(0.12)
Fractional logit, percentage points	0.79***	0.46**
	(0.28)	(0.19)
Outside-node-set share within 100 km	1.11***	0.55***
	(0.30)	(0.16)
Outside-node-set share within 300 km	0.97**	0.48***
	(0.33)	(0.13)

Note: The 100 largest metropolitan areas, benchmark network, 41 annual networks 1985–2025 unless the row names other annual networks. Each row is the WRLURI 2006 coefficient of the design of Table D.3, Panel A, with the change the row names, and the first row is that design itself. Column (1) is its column (2), with Census-division effects on all 100 areas, and column (2) is its column (3), with principal-state effects on the 64 areas in principal states with at least three sample areas. An area's principal state is the state holding the largest share of its 2010 housing units. Rows with the log outcome are 100 times the coefficient, approximately the percent difference in external connectedness per standard deviation of WRLURI 2006. The level and fractional-logit rows are in percentage points of external connectedness, the fractional row as an average partial association. The regression without the two population controls is Table D.3, column (1). Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws). The fractional-logit model is not linear, so its row reports the standard deviation of 999 metropolitan-area pairs-bootstrap draws, with stars from a t distribution with 99 degrees of freedom in column (1) and 63 in column (2).

OA.4.3 Where regulated metropolitan areas draw their dependence from

The metropolitan system can ask a question the state system cannot pose at all. When a regulated market is externally connected, is it connected to its neighbors inside the same state, or to markets elsewhere? The two answers carry opposite implications. Connectedness concentrated among same-state metropolitan areas would mean the state-scale results largely record tight internal clusters, and that aggregating a state's markets into one index measures composition rather than dependence. Connectedness running to markets in other states would mean the reverse.

Table OA.19 splits each metropolitan area's external connectedness by the principal state of the source, same state or other states, and re-runs the regression on the log of each component. Two features of the decomposition govern how it must be read. Row normalization pins the total near 100, so the total varies little across areas (its within-year cross-sectional standard deviation is 4.6 points) while the two components vary by 14.6 and 14.8 among the areas that have both. Almost all of the cross-sectional variation is in the split. And that split depends in large part on how many sample metropolitan areas happen to share an area's state, a feature of the node set rather than of the housing market. Only principal-state effects absorb it, and the table reports those columns. Eighteen of the 100 areas have no same-state sample partner, so the component regressions use the other 82 and the total regression all 100.

In the raw network a one-standard-deviation increase in regulation is associated with 3.9 percent more connectedness to markets in other states and 12.4 percent less to markets within the same state (column 1). Regulated areas therefore take only slightly more of their forecast-error variance from elsewhere in total, but much more from other states and less from their own. Beyond the national cycle the other-state association is 4.1 percent across all areas and 5.0 percent among the 64 areas in states with at least three sample metropolitan areas, and the same-state one -8.2 and -10.6 (columns 2 and 4). The split is by state, not by distance, and many nearby partners, such as New York and Bridgeport or Boston and Providence, count as other states.

Table OA.20 reports every coefficient of these regressions in both networks, including the belt- and division-effect columns, and repeats the two component regressions with the components in percentage points. Within states, a standard deviation of regulation is associated with 2.5 points more connectedness to other states' markets and 2.1 fewer within the state across the 82 areas with a same-state partner in the raw network, and with 2.9 and -2.4 among the 64; with belt or division effects neither component is distinguishable from zero. Among the 64 areas the purge changes neither component by an amount distinguishable from zero; the same-state coefficient is -10.63 (3.63) against -13.78 (3.93) raw, and the other-state coefficient 4.97 (1.78) against 4.50 (1.02). Across the 82 areas with principal-state effects the purge moves the same-state coefficient toward zero by 4.2 (1.5), so part of the same-state shortfall runs through the national factor. The paired differences are also distinguishable from zero in the division column, where neither network's component coefficient is.

Table OA.19: External connectedness of metropolitan areas split by the source's state, raw and factor-adjusted networks, 1985–2025

	All areas		States with ≥ 3 metros	
	Raw	Factor-adjusted	Raw	Factor-adjusted
	(1)	(2)	(3)	(4)
WRLURI 2006 coefficient, dependent variable the log of the row's outcome				
External connectedness, total	0.45*	0.14	0.52***	0.35
	(0.19)	(0.31)	(0.14)	(0.29)
Component from same-state metros	−12.36***	−8.18*	−13.78***	−10.63***
	(3.77)	(3.66)	(3.93)	(3.63)
Component from other states' metros	3.92***	4.08*	4.50***	4.97**
	(1.06)	(1.73)	(1.02)	(1.78)
Controls of Table D.3	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Principal-state fixed effects	✓	✓	✓	✓
Areas, total row / component rows	100 / 82	100 / 82	64 / 64	64 / 64
Sample	All areas	All areas	≥ 3 metros	≥ 3 metros

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025. Columns (1) and (3) use the raw metropolitan benchmark network and columns (2) and (4) the factor-adjusted one. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Each area's external connectedness is split by the principal state of the source (the state holding the largest share of the source area's 2010 housing units) into the part from areas of the same principal state and the part from areas of other states; the two components sum to the total. The split is by state, not by distance. Each cell is the WRLURI coefficient of its own regression of the log of the row's dependent variable on the regressors, year effects and principal-state effects of Appendix Table D.3, column 3, estimated on all areas (columns 1–2) or on the areas of the principal states with at least three sample areas (columns 3–4). The controls are the steep land share, lagged log population and the outside-node-set share. The 18 areas with no same-state sample partner have a same-state component that is zero by design, so in columns (1) and (2) the component regressions use the other 82 areas and the total regressions all 100; in columns (3) and (4) all three use the same 64 areas. Among the 82, the same-state component averages 24.48 and the other-state component 67.38 percentage points in the raw network, and 32.58 and 54.83 in the factor-adjusted network. The belt- and division-effect columns, the level (percentage-point) versions and every coefficient of the raw network are in Table OA.20. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome; on a logged control, it is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

The within-state comparison leans on a few areas. Twenty-one of the 100 areas have counties in more than one state, and they include several of the most regulated, among them Boston, Providence, Philadelphia, New York and Washington, D.C. Their nearest partners count as other states, while their same-state partners, upstate New York or Richmond and Norfolk, are distant, so the split can reflect where state lines fall. Table OA.21 holds this fixed in two ways. With the share of an area's 2010 housing units outside its principal state added, the raw coefficients are −9.8 percent for the same-state component and 3.3 for the other-state one, and the factor-adjusted ones −5.4 and 3.1, with wild bootstrap p-values of 0.11 and 0.10. With the areas in more than one state dropped, the raw coefficients are −7.2 and 3.5 and the factor-adjusted ones −4.5 and 3.4, with p-values of 0.05 and 0.12. The split keeps its sign under both checks in both networks and stays precise in the raw

network. Beyond the national cycle, between a third and a half of the same-state coefficient and up to a quarter of the other-state one rest on the areas that straddle state lines.

Table OA.20: The destination-split regressions in full, raw and factor-adjusted networks, 1985–2025

	(1) Belt		(2) Division		(3) State		(4) State, ≥ 3 metros	
	Raw	Factor-adjusted	Raw	Factor-adjusted	Raw	Factor-adjusted	Raw	Factor-adjusted
Log total								
WRLURI 2006	0.70	0.64	0.88**	0.71	0.45*	0.14	0.52***	0.35
	(0.40)	(0.67)	(0.31)	(0.65)	(0.19)	(0.31)	(0.14)	(0.29)
Factor-adjusted minus raw	−0.06		−0.17		−0.30		−0.17	
	(0.40)		(0.45)		(0.25)		(0.26)	
Steep land share	0.16	0.85*	0.18	0.01	−0.30	−0.09	−0.23	0.06
	(0.23)	(0.45)	(0.43)	(0.71)	(0.16)	(0.24)	(0.16)	(0.23)
Lagged log population	0.05	−0.79	−0.05	−0.91	−0.08	0.04	0.07	0.24
	(0.26)	(0.62)	(0.26)	(0.61)	(0.15)	(0.21)	(0.13)	(0.23)
Outside-node-set share	−1.42***	−1.91**	−1.60***	−1.74***	−0.84***	0.02	−0.71***	−0.10
	(0.42)	(0.70)	(0.31)	(0.57)	(0.24)	(0.24)	(0.23)	(0.23)
Log same-state component								
WRLURI 2006	−9.98	−7.13	−2.18	3.06	−12.36***	−8.18*	−13.78***	−10.63***
	(9.61)	(8.87)	(8.04)	(6.89)	(3.77)	(3.66)	(3.93)	(3.63)
Factor-adjusted minus raw	2.86		5.24**		4.17**		3.16	
	(2.21)		(2.17)		(1.49)		(1.57)	
Steep land share	14.27*	6.70	−2.21	−3.40	2.17	−0.71	1.14	−2.20
	(7.34)	(5.89)	(8.56)	(7.01)	(2.78)	(2.69)	(3.09)	(3.12)
Lagged log population	−13.54	−13.87	−15.58	−15.57*	−6.34	−6.74*	−8.63*	−9.28**
	(9.97)	(8.55)	(8.62)	(7.50)	(3.18)	(2.88)	(3.58)	(3.06)
Outside-node-set share	−10.08	−10.11	−20.16	−23.34**	11.71***	3.11	11.22**	3.75
	(11.33)	(9.85)	(11.71)	(9.43)	(3.66)	(2.28)	(4.09)	(2.56)
Log other-state component								
WRLURI 2006	2.15	2.73	0.05	−1.62	3.92***	4.08*	4.50***	4.97**
	(2.38)	(3.05)	(1.94)	(2.31)	(1.06)	(1.73)	(1.02)	(1.78)
Factor-adjusted minus raw	0.57		−1.67		0.16		0.46	
	(1.20)		(0.91)		(1.20)		(1.30)	
Steep land share	−5.91**	−3.20	0.78	1.89	−0.76	0.43	−0.38	0.96
	(2.04)	(2.15)	(2.70)	(2.62)	(1.14)	(1.46)	(1.24)	(1.60)
Lagged log population	2.63	3.64	4.20	5.48*	1.63	2.20	2.53*	3.17*
	(2.67)	(3.18)	(2.22)	(2.73)	(0.96)	(1.20)	(0.99)	(1.27)
Outside-node-set share	2.98	4.42	5.32	9.43**	−4.74**	−2.02	−4.76**	−2.51*
	(3.16)	(3.67)	(3.36)	(3.56)	(1.53)	(1.19)	(1.70)	(1.21)
Levels (pp), WRLURI 2006 coefficient								
Same-state component	−1.11	−1.16	0.32	1.20	−2.10***	−2.03**	−2.41***	−2.41***
	(1.65)	(2.01)	(1.35)	(1.55)	(0.59)	(0.79)	(0.60)	(0.83)
Factor-adjusted minus raw	−0.05		0.88**		0.07		0.00	
	(0.56)		(0.38)		(0.42)		(0.46)	
Other-state component	1.66	1.69	0.37	−0.70	2.52***	2.16*	2.89***	2.72**
	(1.48)	(1.68)	(1.24)	(1.26)	(0.71)	(0.93)	(0.70)	(0.93)
Factor-adjusted minus raw	0.04		−1.07*		−0.36		−0.17	
	(0.63)		(0.48)		(0.55)		(0.60)	
Fixed effects beyond year	Belt		Division		State		State	
Metropolitan areas total; components	100; 82		100; 82		100; 82		64; 64	

Note: The table reports the regressions of Table OA.19 with every coefficient and with the belt- and division-effect columns, over the 100 metropolitan areas in the 41 annual networks 1985–2025. In each pair of columns the first uses the raw 100-metro benchmark network and the second the factor-adjusted one, and covariates and samples are identical. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The dependent variables are the log of external connectedness (FROM) in total and its same-state and other-state components, and the last panel reports the two components in levels. Log rows are 100 times the coefficient, and 100 times an elasticity on lagged log population. The level rows are in percentage points. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws). Each difference row is the factor-adjusted minus the raw WRLURI coefficient, from one regression stacking both networks and clustered by area. Standard errors hold each estimated network fixed.

Table OA.21: The destination split and metropolitan areas that straddle a state line, raw and factor-adjusted networks, 1985–2025

	Raw			Factor-adjusted		
	(1)	(2)	(3)	(4)	(5)	(6)
Coefficient on WRLURI 2006, with the log of each row's component as the dependent variable						
Panel A. Areas with a same-state sample partner						
Component from same-state metros	−12.36***	−9.77***	−7.18***	−8.18*	−5.35	−4.53**
	(3.77)	(2.78)	(1.99)	(3.66)	(2.59)	(1.83)
Component from other states' metros	3.92***	3.32***	3.45***	4.08*	3.05*	3.44
	(1.06)	(0.88)	(1.00)	(1.73)	(1.49)	(1.70)
Metropolitan areas	82	82	67	82	82	67
Panel B. States with at least three sample areas						
Component from same-state metros	−13.78***	−9.10**	−6.20***	−10.63***	−6.07	−3.98
	(3.93)	(3.12)	(2.05)	(3.63)	(2.86)	(1.96)
Component from other states' metros	4.50***	3.65***	3.53***	4.97**	3.37	3.33
	(1.02)	(0.99)	(1.09)	(1.78)	(1.66)	(1.88)
Metropolitan areas	64	64	52	64	64	52
Share of housing outside the principal state	–	✓	–	–	✓	–
Areas in more than one state dropped	–	–	✓	–	–	✓
Area controls	✓	✓	✓	✓	✓	✓
Year and principal-state fixed effects	✓	✓	✓	✓	✓	✓

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025. Columns (1) and (4) are the regressions of Table OA.19. Each dependent variable is the log of one component of the area's external connectedness (FROM), the part received from sample metropolitan areas in its own principal state or the part received from those in other states. Columns (2) and (5) add the share of the area's 2010 housing units that lies outside its principal state, standardized across the 100 areas. Columns (3) and (6) drop the areas whose counties lie in more than one state, keeping the samples otherwise as defined. Within-state comparisons then rest on the remaining areas of each principal state. The area controls of every column are the steep land share, lagged log population and the outside-node-set share, and every column has year and principal-state fixed effects. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

OA.5 Migration and the Allocation of Connectedness

The tables of this section that extend Table 4, the target-supply interactions, the own-lag networks, the direct and common-origin parts of exposure and the radius exclusions, use its geography controls, in which log distance, contiguity (a shared Census division at metropolitan scale) and same Census region enter with their pairwise interactions. The other tables, the exposure constructions, the gravity benchmarks, the least squares and Poisson comparison, the migration pools and the held-out endpoints, enter the three controls without their interactions.

OA.5.1 Metropolitan exposure constructions and same-state pairs

Table OA.24 re-estimates the metropolitan allocation regressions under three pool rules and both normalizations. The exposure coefficient is insensitive to the construction. Across the six variants the saturated column ranges from 0.018 to 0.027, and the rolling and fixed pools correlate at 0.99 across endpoints. In the factor-adjusted network (Panel B) the designs without pair effects give coefficients two to three and a half times as large under every construction, 0.425 (0.072) against 0.185 (0.033) under source-by-year and target-by-year effects with the rolling within-system pool. The saturated column ranges from 0.025 to 0.042 and is less precise, with shared-node p-values from 0.048 to 0.128.

The sensitivity that matters is to pairs of metropolitan areas with the same principal state, which the state panel does not contain. They are 3.7 percent of dyads, carry 19 percent of the variance of exposure (mean 5.5 percentage points against -0.2 for other pairs) and 39 percent of the variance of conditional shares. Table 4 therefore reports the metropolitan estimates with and without them. In the raw network the saturated coefficient of 0.019 across all pairs (Table OA.23, column 5) is a mixture. The same-state slope is 0.028 (0.010), and restricting the panel to out-of-state pairs leaves 0.008 (0.009), an estimate as small and as indistinguishable from zero as the state panel's -0.010 (0.018). Metropolitan areas in the same state share state-level shocks, institutions and lenders as well as migrants, and the allocation design cannot separate those channels for in-state dyads.

The same channel carries the heterogeneity by regulation. Table OA.22 interacts conditional migration exposure with the target's standardized regulation and steep land share in column (4) of Table 4, with source-by-year and target-by-year effects and that table's interacted geography controls. The direct weights of Section 5 imply no interaction in conditional shares, because the supply multiplier $\rho_{i,t}$ scales every weight of a target's row and cancels from conditional weights, so the table asks whether allocation departs from that prediction. Among metropolitan pairs in different states it does not. The regulation interaction is 0.007 (0.015) in the factor-adjusted network and 0.003 (0.007) in the raw one, the slope interaction 0.003 (0.013) and 0.000 (0.006), and on the apportioned node set of Table 8 the factor-adjusted interactions are -0.004 (0.010) and -0.003 (0.008). Across states they are -0.001 (0.008) and 0.019 (0.012) in the factor-adjusted network and 0.015 (0.009) and 0.007 (0.011) in the raw one.

Across all metropolitan pairs the regulation interaction is negative, −0.063 (0.021) in the factor-adjusted network and −0.038 (0.009) in the raw one, and −0.057 (0.035) on the apportioned node set. Read on its own, that says regulated markets follow their migration linkages less closely. The destination split of Section OA.4.3 offers a reading of it. Same-state pairs are the pairs with the highest exposure, and regulated metropolitan areas place less of their dependence on exactly those partners and more on markets in other states. Over all pairs, therefore, the high-exposure same-state partners of a regulated target receive less connectedness than exposure alone would predict, and the exposure gradient flattens with regulation. The pattern is consistent with in-state clusters rather than with migration-linked sources in general, a conjecture rather than an estimate, and it vanishes once same-state pairs are excluded.

Table OA.22: Allocation and the target's supply constraint, column (4) of Table 4, 2005–2025

		Benchmark, 100 largest		Apportioned, 100	
	States	All pairs	Out of state	All pairs	Out of state
	(1)	(2)	(3)	(4)	(5)
Panel A. Raw networks					
Exposure, Table 4 column (4)	−0.014	0.060***	0.009	0.067***	0.018***
	(0.020)	(0.022)	(0.010)	(0.018)	(0.007)
Exposure	−0.016	0.063***	0.008	0.067***	0.017***
	(0.019)	(0.020)	(0.010)	(0.019)	(0.006)
Exposure × target WRLURI	0.015	−0.038***	0.003	−0.023*	0.003
	(0.009)	(0.009)	(0.007)	(0.013)	(0.004)
Exposure × target steep land share	0.007	0.004	0.000	−0.002	0.005
	(0.011)	(0.009)	(0.006)	(0.012)	(0.003)
Pair-years	47,376	207,900	200,256	205,821	202,125
Panel B. Factor-adjusted networks					
Exposure, Table 4 column (4)	0.000	0.171***	0.060**	0.204***	0.053***
	(0.016)	(0.056)	(0.025)	(0.047)	(0.013)
Exposure	−0.002	0.176***	0.059**	0.202***	0.054***
	(0.011)	(0.052)	(0.025)	(0.045)	(0.013)
Exposure × target WRLURI	−0.001	−0.063***	0.007	−0.057*	−0.004
	(0.008)	(0.021)	(0.015)	(0.035)	(0.010)
Exposure × target steep land share	0.019	0.002	0.003	−0.015	−0.003
	(0.012)	(0.019)	(0.013)	(0.030)	(0.008)
Pair-years	47,376	207,900	200,256	205,821	202,125

Note: Dependent variable: the conditional connectedness share (percentage points), 21 annual networks 2005–2025, each node set's own networks and migration exposure. Every column has source-by-year and target-by-year effects and the interacted geography controls of Table 4. The first row of each panel is the exposure coefficient of column (4) of that table without interactions. The next three rows are one regression that adds exposure times the target's WRLURI 2006 and steep land share, each standardized over the node set's targets, whose main effects the target-by-year effects absorb. The model predicts no interaction, since the supply multiplier cancels from conditional shares (Section 5). The apportioned node set is that of Table 8. Cheyenne, WY, has no WRLURI respondent, so columns (4) and (5) drop it as a target and keep it as a source; with it, the factor-adjusted coefficients of the first row are 0.201 and 0.052. Parentheses contain shared-node standard errors, which allow two directed pairs to be dependent whenever they share a node in either role. *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against a Student t distribution with one degree of freedom fewer than the number of nodes. The regressions run in Stata (reghdfe, with the shared-node covariance in Mata).

Table OA.23 reports the full set of allocation specifications at both scales in the raw and the factor-adjusted networks, with the three geography controls (log distance, contiguity or a shared Census division, and same Census region) entered without interactions, including the target-supply interactions under unordered-pair-and-year effects. Under those effects the target's own regulation and steep land share vary within a pair because the two directions of a pair have different targets, so the column includes them as main effects. No interaction is then distinguishable from zero at either scale in the raw network. The regulation interaction is −0.022 (0.019) across states and −0.008 (0.006) and −0.006 (0.009) across all and out-of-state metropolitan pairs, and the slope interaction 0.007 (0.006), 0.007 (0.005) and 0.002 (0.006).

Table OA.23: Migration exposure, target supply and the allocation of connectedness across sources, raw and factor-adjusted networks, 2005–2025

Dependent variable	Conditional connectedness share (percentage points)				
	(1)	(2)	(3)	(4)	(5)
Raw networks					
States (48 targets, 47 sources)					
Migration exposure	0.104**	0.093**	0.079**	−0.008	−0.010
	(0.042)	(0.044)	(0.039)	(0.020)	(0.018)
	[0.054]	[0.053]	[0.051]	[0.027]	[0.023]
× target WRLURI 2006			−0.022	0.020**	
			(0.019)	(0.008)	
			[0.026]	[0.011]	
× target steep land share			0.007	0.003	
			(0.006)	(0.009)	
			[0.013]	[0.015]	
Metropolitan areas, all pairs (100 targets, 99 sources)					
Migration exposure	0.107***	0.035**	0.027*	0.104***	0.019***
	(0.033)	(0.017)	(0.016)	(0.026)	(0.006)
	[0.043]	[0.023]	[0.021]	[0.035]	[0.008]
× target WRLURI 2006			−0.008	−0.036***	
			(0.006)	(0.009)	
			[0.008]	[0.015]	
× target steep land share			0.007	−0.005	
			(0.005)	(0.010)	
			[0.007]	[0.014]	
Metropolitan areas, out-of-state pairs					
Migration exposure	0.052**	0.041	0.026	0.016	0.008
	(0.026)	(0.035)	(0.031)	(0.011)	(0.009)
	[0.035]	[0.049]	[0.046]	[0.015]	[0.014]
× target WRLURI 2006			−0.006	0.003	
			(0.009)	(0.007)	
			[0.014]	[0.010]	
× target steep land share			0.002	−0.004	
			(0.006)	(0.006)	
			[0.012]	[0.009]	
Geography	✓	–	–	✓	–
Year fixed effects	✓	✓	✓	–	–
Unordered-pair fixed effects	–	✓	✓	–	✓
Source-by-year and target-by-year fixed effects	–	–	–	✓	✓
Target main effects	–	–	✓	–	–
Directed pair-years, states	47,376	47,376	47,376	47,376	47,376
metropolitan, all pairs	207,900	207,900	207,900	207,900	207,900
metropolitan, out-of-state pairs	200,256	200,256	200,256	200,256	200,256
Within R^2					
States	0.0348	0.0092	0.0233	0.0479	0.0001
Metropolitan, all pairs	0.2048	0.0046	0.0402	0.2803	0.0014
Metropolitan, out-of-state pairs	0.0327	0.0048	0.0555	0.0452	0.0002

Note: 21 annual benchmark networks 2005–2025. This part reports the raw network and the continuation the factor-adjusted one. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The dependent variable, the regressor, the geography controls, the samples and the shared-node standard errors and stars are those of Table 4; brackets contain delete-one-node jackknife standard errors. Column (1) has geography and year effects, column (2) unordered-pair-and-year effects, and column (5) source-by-year, target-by-year and unordered-pair effects, the saturated design. Columns (3) and (4) add the interactions of conditional migration exposure with the target's standardized WRLURI 2006 and steep land share, under unordered-pair-and-year effects in column (3) and under node-year effects with geography in column (4). Conditional shares average $100/47 = 2.13$ percentage points across states, $100/99 = 1.01$ across all metropolitan pairs and 0.88 across out-of-state pairs. Unordered-pair effects do not absorb a target characteristic, so column (3) includes the two target main effects, given here to four decimals with shared-node standard errors. They are WRLURI 2006 −0.2705 (0.0834) and steep land share −0.0648 (0.0903) across states, WRLURI 2006 −0.1354 (0.0271) and steep land share −0.1029 (0.0353) across all metropolitan pairs and WRLURI 2006 −0.1289 (0.0271) and steep land share −0.1111 (0.0364) across out-of-state pairs. In column (4) the target-by-year effects absorb them.

Table OA.23 (continued). Factor-adjusted networks

Dependent variable	Conditional connectedness share (percentage points)				
	(1)	(2)	(3)	(4)	(5)
Factor-adjusted networks					
States (48 targets, 47 sources)					
Migration exposure	0.044*	0.026	0.015	0.009	−0.003
	(0.024)	(0.023)	(0.026)	(0.012)	(0.017)
	[0.030]	[0.029]	[0.036]	[0.020]	[0.023]
× target WRLURI 2006			−0.020	0.005	
			(0.016)	(0.007)	
			[0.024]	[0.012]	
× target steep land share			0.007	0.016	
			(0.011)	(0.011)	
			[0.019]	[0.019]	
Metropolitan areas, all pairs (100 targets, 99 sources)					
Migration exposure	0.232***	0.035**	0.028**	0.277***	0.025*
	(0.059)	(0.013)	(0.013)	(0.064)	(0.014)
	[0.078]	[0.017]	[0.017]	[0.088]	[0.018]
× target WRLURI 2006			0.014*	−0.057**	
			(0.008)	(0.023)	
			[0.013]	[0.042]	
× target steep land share			−0.006	−0.021	
			(0.008)	(0.021)	
			[0.011]	[0.034]	
Metropolitan areas, out-of-state pairs					
Migration exposure	0.074***	0.035*	0.026	0.069**	0.011
	(0.021)	(0.019)	(0.017)	(0.026)	(0.014)
	[0.030]	[0.025]	[0.025]	[0.042]	[0.021]
× target WRLURI 2006			0.011	0.008	
			(0.011)	(0.016)	
			[0.018]	[0.025]	
× target steep land share			−0.002	−0.002	
			(0.007)	(0.012)	
			[0.011]	[0.021]	
Geography	✓	–	–	✓	–
Year fixed effects	✓	✓	✓	–	–
Unordered-pair fixed effects	–	✓	✓	–	✓
Source-by-year and target-by-year fixed effects	–	–	–	✓	✓
Target main effects	–	–	✓	–	–
Directed pair-years, states	47,376	47,376	47,376	47,376	47,376
metropolitan, all pairs	207,900	207,900	207,900	207,900	207,900
metropolitan, out-of-state pairs	200,256	200,256	200,256	200,256	200,256
Within R^2					
States	0.0101	0.0005	0.0071	0.0127	0.0000
Metropolitan, all pairs	0.3012	0.0034	0.0112	0.3427	0.0013
Metropolitan, out-of-state pairs	0.0309	0.0030	0.0145	0.0259	0.0002

Note: Factor-adjusted benchmark networks; construction, samples and inference as in the first part of the table. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Factor-adjusted conditional shares average $100/47 = 2.13$ percentage points across states, $100/99 = 1.01$ across all metropolitan pairs and 0.71 across out-of-state pairs. Unordered-pair effects do not absorb a target characteristic, so column (3) includes the two target main effects, given here to four decimals with shared-node standard errors. They are WRLURI 2006 −0.2211 (0.0678) and steep land share −0.0496 (0.0689) across states, WRLURI 2006 −0.0785 (0.0309) and steep land share −0.0389 (0.0276) across all metropolitan pairs and WRLURI 2006 −0.0777 (0.0309) and steep land share −0.0395 (0.0282) across out-of-state pairs. In column (4) the target-by-year effects absorb them. The factor-adjusted main effects of the two metropolitan samples agree to three decimals, although each comes from its own regression (207,900 and 200,256 directed pair-years).

Table OA.24: Metropolitan source allocation across exposure constructions, raw and factor-adjusted networks, 2005–2025

Dependent variable:	Conditional connectedness share (pp)				
Pool, normalization	(1)	(2)	(3)	(4)	(5)
Panel A. Raw network					
Fixed 1990–1994, outside (21 endpoints)	0.127***	0.052*	0.242***	0.121***	0.027***
	(0.044)	(0.027)	(0.057)	(0.045)	(0.009)
Fixed 1990–1994, within (21 endpoints)	0.111***	0.040**	0.189***	0.102***	0.022***
	(0.033)	(0.018)	(0.035)	(0.030)	(0.006)
Rolling $t-15$ to $t-11$, outside (21 endpoints)	0.123***	0.045*	0.238***	0.121***	0.022**
	(0.044)	(0.026)	(0.054)	(0.043)	(0.008)
Rolling $t-15$ to $t-11$, within (21 endpoints)	0.107***	0.035**	0.185***	0.101***	0.019***
	(0.033)	(0.017)	(0.033)	(0.029)	(0.006)
Rolling $t-20$ to $t-16$, outside (16 endpoints)	0.105***	0.040	0.201***	0.096***	0.020***
	(0.040)	(0.025)	(0.047)	(0.036)	(0.006)
Rolling $t-20$ to $t-16$, within (16 endpoints)	0.090***	0.031*	0.156***	0.080***	0.018***
	(0.029)	(0.016)	(0.028)	(0.024)	(0.005)
Panel B. Factor-adjusted network					
Fixed 1990–1994, outside (21 endpoints)	0.260***	0.047**	0.555***	0.329***	0.036
	(0.073)	(0.020)	(0.123)	(0.106)	(0.024)
Fixed 1990–1994, within (21 endpoints)	0.235***	0.036**	0.431***	0.274***	0.025*
	(0.059)	(0.014)	(0.075)	(0.072)	(0.014)
Rolling $t-15$ to $t-11$, outside (21 endpoints)	0.260***	0.048**	0.550***	0.335***	0.042*
	(0.075)	(0.020)	(0.119)	(0.105)	(0.023)
Rolling $t-15$ to $t-11$, within (21 endpoints)	0.232***	0.035**	0.425***	0.274***	0.025*
	(0.059)	(0.013)	(0.072)	(0.070)	(0.014)
Rolling $t-20$ to $t-16$, outside (16 endpoints)	0.264***	0.047**	0.561***	0.338***	0.042*
	(0.076)	(0.021)	(0.121)	(0.106)	(0.025)
Rolling $t-20$ to $t-16$, within (16 endpoints)	0.237***	0.037**	0.435***	0.278***	0.029**
	(0.060)	(0.015)	(0.072)	(0.070)	(0.014)
Geography	✓	–	–	✓	–
Year fixed effects	✓	✓	–	–	–
Unordered-pair fixed effects	–	✓	–	–	✓
Source-by-year and target-by-year fixed effects	–	–	✓	✓	✓

Note: The table re-estimates the metropolitan allocation regressions under alternative exposure constructions. The dependent variable and regressor are those of Table 4. Panel A uses the raw 100-metro benchmark network and Panel B the factor-adjusted one, and exposures, geography, samples and fixed effects are identical. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Entries are coefficients on conditional migration exposure over all 9,900 directed metropolitan pairs in the annual networks 2005–2025, with shared-node standard errors in parentheses. Stars come from t_{99}, and *, ** and *** denote $p < 0.10$, 0.05 and 0.01. Each pool rule pools IRS flow-label years over the stated window before endpoint t. The $t-15$ to $t-11$ rule is strictly predetermined at the 120-month window and feasible at all 21 endpoints, and the $t-20$ to $t-16$ rule loses 2005–2009. The *within* normalization normalizes flows over the 100-metro system, and the *outside* normalization routes flows with one end outside the system through an absorbing node before the two-step product. Columns (1) to (5) carry geography and year effects, unordered-pair-and-year effects, source-by-year and target-by-year effects, node-year effects with geography, and node-year plus unordered-pair effects (saturated). Standard errors hold each estimated network fixed.

OA.5.2 Gravity benchmarks

Table OA.25 compares gravity and unordered-pair specifications at state scale, including models without migration exposure, in the raw network (Panel A) and the factor-adjusted one (Panel B). The fit increment measures how much adding exposure improves fit with the same controls. The rows with unordered-pair and source-by-year effects and with unordered-pair and target-by-year effects isolate the contribution of each set of node-year effects, and the main allocation table includes both sets together. Migration's association with the allocation is less than half as large in the factor-adjusted network, 0.044 (0.024) with geography against 0.104 (0.042), and 0.026 (0.023) with unordered-pair-and-year effects against 0.093 (0.044). The incremental fit of exposure over geography falls from 0.020 to 0.003. With the interacted geography controls the two geography slopes are 0.104 (0.044) and 0.039 (0.025) (Table OA.30, column 1).

Table OA.25: Geography benchmarks and the incremental association of migration exposure, raw and factor-adjusted networks, 2005–2025

Design	Estimator	Exposure	Shared-node SE	Node JK SE	Fit	Incremental fit
Panel A. Raw network						
Geography and year, no exposure	OLS	–	–	–	0.0147	–
Geography and year	OLS	0.104	0.042	0.054	0.0348	0.0201
Unordered-pair and year, no exposure	OLS	–	–	–	0.0000	–
Unordered-pair and year	OLS	0.093	0.044	0.053	0.0092	0.0092
Unordered-pair and source-by-year	OLS	−0.010	0.018	0.023	0.0001	0.0001
Unordered-pair and target-by-year	OLS	−0.010	0.018	0.023	0.0001	0.0001
Geography and year, no exposure	PPML	–	–	–	0.0062	–
Geography and year	PPML	0.076	0.026	0.034	0.0131	0.0069
Unordered-pair and year, no exposure	PPML	–	–	–	0.0991	–
Unordered-pair and year	PPML	0.073	0.031	0.037	0.1015	0.0024
Unordered-pair and source-by-year	PPML	−0.018	0.009	0.013	0.2659	0.0001
Unordered-pair and target-by-year	PPML	−0.018	0.010	0.013	0.1463	0.0001
Panel B. Factor-adjusted network						
Geography and year, no exposure	OLS	–	–	–	0.0074	–
Geography and year	OLS	0.044	0.024	0.030	0.0101	0.0027
Unordered-pair and year, no exposure	OLS	–	–	–	0.0000	–
Unordered-pair and year	OLS	0.026	0.023	0.029	0.0005	0.0005
Unordered-pair and source-by-year	OLS	−0.003	0.017	0.023	0.0000	0.0000
Unordered-pair and target-by-year	OLS	−0.002	0.017	0.024	0.0000	0.0000
Geography and year, no exposure	PPML	–	–	–	0.0039	–
Geography and year	PPML	0.036	0.018	0.022	0.0052	0.0013
Unordered-pair and year, no exposure	PPML	–	–	–	0.0939	–
Unordered-pair and year	PPML	0.023	0.021	0.027	0.0941	0.0002
Unordered-pair and source-by-year	PPML	−0.004	0.013	0.019	0.1551	0.0000
Unordered-pair and target-by-year	PPML	−0.003	0.014	0.019	0.1093	0.0000

Note: The table reports geography and unordered-pair specifications of the state allocation regression, with and without conditional migration exposure. Panel A uses the raw state benchmark network and Panel B the factor-adjusted one, and exposure, geography, samples and fixed effects are identical. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The sample has 48 targets and 47 sources in the 21 annual networks 2005–2025 (47,376 directed pair-years). Conditional connectedness and the conditional migration share (uncentered) are percentage-point shares of a target's total over its 47 external sources, and the regressor, conditional migration exposure, is that share minus 100/47. Exposure is built from IRS flows pooled over $t-15$ through $t-11$. Geography is log distance, contiguity and same Census region, and pair effects are unordered. Rows name the geography and fixed effects of each design, and rows marked no exposure omit the regressor. PPML entries are average partial associations, the index coefficient times 100/47. Fit is the within R^2 for OLS and the pseudo-R^2 for PPML, and incremental fit compares models with the same fixed effects and geography. Fit differences are descriptive. The standard-error columns report shared-node and delete-one-node jackknife standard errors. Standard errors hold each estimated network fixed.

OA.5.3 Remaining migration variation

Table OA.26 reports the exposure variation left by each set of fixed effects and its partial fit with conditional connectedness. These diagnostics use the same regressor and sample as the allocation estimates. The partial R^2 measures the squared correlation of the outcome and exposure after removing the listed effects; it is not a power calculation.

Table OA.26: Remaining migration variation and partial model fit

Effects removed	Variance retained (%)	Residual SD	Partial R^2 (%)
Year and pair	26.44	1.729	0.925
Source-year and pair	13.65	1.242	0.011
Target-year and pair	13.77	1.247	0.006
Source-year, target-year and pair	13.65	1.242	0.011

Note: All projections use 47,376 observations. Raw exposure variance is 11.30. Adding target-year effects after source-year and unordered-pair effects removes less than 0.001% of the remaining exposure variation. Partial R^2 measures the additional fit from exposure after the listed effects; it is not a power calculation.

OA.5.4 Least squares and Poisson estimates

Table OA.27 pairs the node-year least-squares specifications with PPML, in the raw and the factor-adjusted networks. Because target-by-year effects make fitted external-row totals equal observed totals, the average PPML derivative equals the index slope times the common mean row share, 100/47. The table reports that average partial association on the same share-point scale as OLS. In the raw network the saturated design is −0.010 (0.018) under OLS and −0.018 (0.009) under PPML. The source-by-year and target-by-year design keeps its association in the factor-adjusted network, 0.059 (0.017) under OLS and 0.051 (0.011) under PPML, and the two designs that add geography or pair effects to node-year effects stay near zero, as they do in the raw network.

Table OA.27: Conditional allocation by OLS and PPML with node-based inference, raw and factor-adjusted networks, 2005–2025

	Raw network		Factor-adjusted network	
Model	OLS	PPML	OLS	PPML
Geography and year	0.104**		0.044*	
	(0.042)		(0.024)	
	[0.054]		[0.030]	
Unordered-pair and year	0.093**		0.026	
	(0.044)		(0.023)	
	[0.053]		[0.029]	
Source-by-year and target-by-year	0.068***	0.053***	0.059***	0.051***
	(0.014)	(0.010)	(0.017)	(0.011)
	[0.018]	[0.012]	[0.022]	[0.015]
Node-year and geography	−0.004	−0.012	0.011	0.008
	(0.019)	(0.011)	(0.016)	(0.012)
	[0.024]	[0.015]	[0.023]	[0.018]
Saturated	−0.010	−0.018*	−0.003	−0.003
	(0.018)	(0.009)	(0.017)	(0.013)
	[0.023]	[0.013]	[0.023]	[0.019]

Note: The table estimates the state allocation designs by least squares and, where target-by-year effects fix each row's total, by PPML. The dependent variable and regressor are those of Table 4. Columns 1–2 use the raw state benchmark network and columns 3–4 the factor-adjusted one, and exposure, geography, samples and fixed effects are identical. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The sample has 48 targets and 47 sources in the 21 annual networks 2005–2025 (47,376 directed pair-years). Conditional connectedness and the conditional migration share (uncentered) are percentage-point shares of a target's total over its 47 external sources, and the regressor, conditional migration exposure, is that share minus 100/47. Exposure is built from IRS flows pooled over $t-15$ through $t-11$. Geography is log distance, contiguity and same Census region, and pair effects are unordered. The rows carry geography and year effects, unordered-pair-and-year effects, source-by-year and target-by-year effects, node-year effects with geography, and node-year plus unordered-pair effects (saturated). PPML entries are average partial associations, the index coefficient times 100/47. Parentheses contain shared-node standard errors and brackets delete-one-node jackknife standard errors. Stars use the shared-node t statistic with 47 degrees of freedom, and *, ** and *** denote $p < 0.10$, 0.05 and 0.01. Standard errors hold each estimated network fixed.

OA.5.5 Fixed and rolling migration pools

Table OA.28 fixes migration exposure at its 1990–1994 value for every endpoint and compares it with the rolling five-year pools. Connectedness outcomes, covariates and the 48-state universe are unchanged. This checks whether the allocation results depend on updating migration weights; it does not use a shorter averaging period.

Table OA.28: Conditional allocation: rolling versus fixed migration pools, 2005–2025

Model	Estimator	Rolling	Shared-node SE	Fixed 1990–94	Shared-node SE
Geography and year	OLS	0.104	0.042	0.105	0.039
Unordered-pair and year	OLS	0.093	0.044	0.096	0.041
Source-year and target-year	OLS	0.068	0.014	0.068	0.013
Source-year and target-year	PPML	0.053	0.010	0.052	0.010
Node-year and geography	OLS	−0.004	0.019	−0.005	0.018
Node-year and geography	PPML	−0.012	0.011	−0.014	0.011
Node-year and pair	OLS	−0.010	0.018	−0.010	0.018
Node-year and pair	PPML	−0.018	0.009	−0.019	0.009

Note: Conditional connectedness and migration exposure are in percentage points across a target's 47 external sources, exposure centered at 100/47; 47,376 directed pair-years, IRS pools from $t-15$ through $t-11$ in the rolling columns and the fixed 1990–94 pool in the fixed columns. Geography is log distance, contiguity and same Census region; pair effects are unordered. Shared-node standard errors are in the columns beside each estimate. The fixed design uses the same 1990–94 migration pool at every endpoint.

OA.5.6 The unconditional cell

The direct weights of Section 5 imply an interaction between exposure and the target's supply constraint, but on the unconditional cell rather than the conditional share. Writing $W_{ij,t} = \rho_{i,t}\psi_{ij,t}$ (equation 9), the conditional share divides by the row total and $\rho_{i,t}$ cancels exactly, whereas the unconditional cell keeps it. Regressing $C_{ij,t}$ on diagonal-inclusive exposure, its interaction with target regulation and its interaction with target openness, under source-by-year and target-by-year effects, returns an interaction of 0.325 with a bootstrap standard error of 0.078, from 999 draws that resample states together with every directed pair in which a drawn state appears. That estimate is an accounting identity rather than evidence. Since $C_{ij} = F_i\bar{C}_{ij}/100$, the cross-derivative decomposes into a scaling term, the receiving-side gradient times the allocation slope, and a term in the conditional interaction. On the same 21 endpoints the two terms are 0.025 and 0.299 and sum to 0.324 against the 0.325 estimated directly; the gap, 0.001 with a standard error of 0.008 from the same draws, is approximation error. The log form carries more content. Poisson pseudo-maximum-likelihood estimation of $C_{ij,t}$ on $\log \psi_{ij,t}$ with the same effects gives an elasticity of 0.134 (0.012) against a direct-weight value of exactly one, which no bootstrap draw reaches. Indirect paths through intermediate markets (the Neumann series of Appendix E.6) and the innovation covariance smooth the mapping from W_{ij} to C_{ij}, so an elasticity below one is expected, and the ordering restriction—that unconditional dependence rises with exposure—holds; the paper claims no more than the ordering. Table OA.29 reports each component, with its bootstrap standard error and percentile interval from the same paired node-cluster draws. The components are products and ratios of regression coefficients, not single coefficients, so this accounting keeps its paired bootstrap rather than the wild bootstrap-t of the regression tables. These calculations are on the raw network.

Table OA.29: The product form of the unconditional cell as an accounting identity, 2005–2025

	Estimate	SE	95-percent interval
Panel A. Unconditional cell $C_{ij,t}$ on raw exposure, source-by-year and target-by-year effects			
$\psi_{ij,t}$	1.087	(0.371)	$[0.618, 2.139]$
$\psi_{ij,t} \times R_i$	0.325	(0.078)	$[0.199, 0.501]$
$\psi_{ij,t} \times o_{i,t}$	−3.396	(5.405)	$[-19.856, 1.940]$
Panel B. The two terms of the identity			
Allocation slope, $\bar{C}_{ij,t}$ on $\psi_{ij,t}$	1.026	(0.121)	$[0.800, 1.269]$
Intensity gradient dF/dR (pp per SD of WRLURI)	2.431	(0.500)	$[1.490, 3.505]$
Mean external connectedness $\bar{F}$ (pp)	85.29	(0.30)	$[84.70, 85.86]$
Conditional interaction, $\bar{C}_{ij,t}$ on $\psi_{ij,t} \times R_i$	0.351	(0.091)	$[0.203, 0.555]$
Scaling term, allocation slope $\times\, dF/dR/100$	0.025	(0.006)	$[0.015, 0.037]$
Conditional term, $\bar{F}/100 \times$ conditional interaction	0.299	(0.077)	$[0.173, 0.474]$
Sum of the two terms	0.324	(0.076)	$[0.204, 0.499]$
Direct estimate minus sum	0.001	(0.008)	$[-0.015, 0.016]$
Panel C. Log form, PPML of $C_{ij,t}$ on $\log \psi_{ij,t}$ with the same effects (direct-weight value 1)			
$\log \psi_{ij,t}$	0.134	(0.012)	$[0.109, 0.156]$

Note: 48 states, 21 annual networks 2005–2025, benchmark network; 47,376 directed pair-years. $\psi_{ij,t}$ is raw, diagonal-inclusive migration exposure in unit shares, $o_{i,t} = 1 - \psi_{ii,t}$ is the target's openness and R_i its standardized WRLURI 2006. Because $C_{ij,t} = F_{i,t}\bar{C}_{ij,t}/100$, the interaction of Panel A decomposes into the scaling and conditional terms of Panel B up to approximation error; the exercise is an accounting identity, not a test. The intensity gradient is the least-squares WRLURI coefficient of external connectedness in percentage points on the Table 2 regressors with year effects, on the same 21 annual networks. Parentheses contain standard errors and brackets 95-percent percentile intervals from 999 paired node-cluster bootstrap draws. Each draw resamples the 48 states, a drawn state brings every directed pair in which it is target or source (a state drawn twice enters twice with relabeled identifiers), and the intensity gradient is refit on the drawn states in the same draw. No draw of the log elasticity reaches one.

OA.5.7 Allocation under the own-lag networks

Table OA.30 reports the allocation specifications under the three networks of Section 7.1 in their raw and factor-adjusted forms, on the 21 endpoints from 2005 for the benchmark and the own-lags-free network and on the endpoints at which all three networks are stable for the eight-own-lag comparison. The designs with geography use the interacted geography controls of Table 4. In the raw networks the geography, unordered-pair-and-year and node-year-only coefficients are positive under every network, at 0.104, 0.093 and 0.068 under the benchmark, and distinguishable from zero at the 5-percent level except in the geography and unordered-pair-and-year designs of the own-lags-free network, whose p-values are 0.07 and 0.17 on the 21 endpoints. The two designs that add geography or unordered-pair effects to node-year effects, the second of them the saturated design, range from −0.014 to 0.003 across the raw networks and samples and are not distinguishable from zero. All 21-endpoint allocation comparisons use 47,376 directed pair-year observations and the 16-endpoint comparisons 36,096. Differences between point estimates across the networks are descriptive; the table does not test one estimator against the other.

Table OA.30: Source allocation under the own-lag networks, raw and factor-adjusted

	Benchmark (1)	Own-lags-free (2)	Benchmark (3)	Own-lags-free (4)	Eight-own-lag (5)
Panel A. Raw networks					
Geography and year	0.104**	0.072*	0.097**	0.064*	0.114***
	(0.044)	(0.039)	(0.042)	(0.035)	(0.034)
Unordered-pair and year	0.093**	0.054	0.086**	0.048	0.112***
	(0.044)	(0.039)	(0.042)	(0.034)	(0.032)
Source-by-year and target-by-year	0.068***	0.074***	0.047***	0.053***	0.055***
	(0.014)	(0.013)	(0.011)	(0.011)	(0.015)
Node-year and geography	−0.014	−0.014	−0.008	−0.004	−0.006
	(0.020)	(0.019)	(0.018)	(0.016)	(0.024)
Node-year and unordered-pair	−0.010	−0.011	−0.008	−0.007	0.003
	(0.018)	(0.018)	(0.015)	(0.013)	(0.018)
Annual networks	21	21	16	16	16
Panel B. Factor-adjusted networks					
Geography and year	0.039	0.027	0.041	0.025	0.023
	(0.025)	(0.021)	(0.025)	(0.021)	(0.016)
Unordered-pair and year	0.026	0.020	0.028	0.017	0.016
	(0.023)	(0.018)	(0.023)	(0.018)	(0.015)
Source-by-year and target-by-year	0.059***	0.051***	0.057***	0.049***	0.040**
	(0.017)	(0.016)	(0.017)	(0.016)	(0.015)
Node-year and geography	0.000	−0.012	0.001	−0.012	−0.018
	(0.016)	(0.015)	(0.017)	(0.016)	(0.020)
Node-year and unordered-pair	−0.003	0.001	−0.003	−0.002	−0.009
	(0.017)	(0.015)	(0.017)	(0.015)	(0.016)
Annual networks	21	21	20	20	20
Panel C. Factor-adjusted networks on the raw common stable endpoints					
Geography and year			0.036	0.020	0.022
			(0.025)	(0.022)	(0.019)
Unordered-pair and year			0.024	0.013	0.014
			(0.019)	(0.017)	(0.015)
Source-by-year and target-by-year			0.051***	0.043***	0.034*
			(0.019)	(0.016)	(0.017)
Node-year and geography			0.009	−0.012	−0.011
			(0.019)	(0.017)	(0.021)
Node-year and unordered-pair			0.002	−0.005	−0.008
			(0.016)	(0.014)	(0.014)
Annual networks			16	16	16

Note: 48 states. Each column estimates the five-design conditional-allocation ladder (geography and year effects; unordered-pair-and-year effects; source-by-year and target-by-year effects; node-year effects with geography; node-year and unordered-pair effects, the saturated design), in percentage points across 47 external sources, on one network. Columns 1–2 use the 21 annual networks 2005–2025 and columns 3–5 the years in which all three networks of the representation are stable, 16 raw and 20 factor-adjusted; Panel C re-estimates the factor-adjusted columns 3–5 on the 16 raw common stable endpoints (2010–2025). The geography controls are those of Table 4 across states, log distance, contiguity and same Census region with their pairwise interactions and the interactions of log distance with each indicator. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Parentheses contain shared-node standard errors. *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against a Student t distribution with one degree of freedom fewer than the number of nodes. The own-lags-free and eight-own-lag networks are described in Section 7.1. The regressions run in Stata (reghdfe, with the shared-node covariance in Mata).

OA.5.8 Held-out endpoints and persistence

Table OA.31 compares three models of conditional source allocation in held-out endpoint years. It uses observed held-out migration exposure and therefore is not an ex-ante house-price forecast. Nor is it out-of-sample validation. Each held-out annual window shares 108 of its 120 months with its neighbors, allocations one year apart have a median within-target rank correlation of 0.78 (Table OA.32), and nearly all exposure variation is between links, so a model fitted on the other 20 endpoints predicts the held-out one largely because allocations and exposure persist. The comparison describes in-sample ranking. Holding out blocks of endpoints at least ten years apart would be needed for a statement about generalization.

Table OA.31: Held-out-year prediction of conditional source allocation, 2005–2025

Model	KL divergence	RMSE (pp)	MAE (pp)	Rank correlation	Folds
Gravity only	0.3459	1.8596	1.4104	0.0498	21
Migration only	0.3404	1.8505	1.3982	0.2278	21
Gravity + migration	0.3400	1.8491	1.3967	0.1998	21

Note: Each of 21 folds holds out one annual network. All models use the same 48 targets, with 47 sources per target, and the benchmark network. KL is the Kullback–Leibler divergence of predicted from observed conditional shares; RMSE and MAE are the root mean squared and mean absolute errors in share points; the rank correlation is the within-target Spearman correlation. Lower divergence and error indicate closer predictions; a higher rank correlation indicates a closer ranking. Each held-out window shares 108 of its 120 months with its neighbors and exposure varies almost entirely between links, so the held-out endpoints are close to copies of the training endpoints: differences are descriptive and do not measure out-of-sample predictive validity.

The connectedness rankings change over longer horizons even though adjacent rolling windows share much of their price history. The median within-target Spearman correlation of source rankings is 0.781 one year apart, 0.384 five years apart, 0.118 ten years apart and 0.110 twenty years apart (Table OA.32, Panel A), so a decade separates networks whose source orderings retain little in common. With the national factor removed the orderings persist less still, at 0.651 one year apart and 0.140 five years apart (Panel B). Migration allocation is more persistent, so the unordered-pair-and-year and saturated designs draw on different amounts of residual variation.

Table OA.32: Persistence of bilateral connectedness allocations, raw and factor-adjusted networks, 2005–2025

Gap	Median ρ_S	Median r	Top 10%	Top 20%	Between share
Panel A. Raw network					
1 year	0.781	0.811	0.596	0.676	0.384
5 years	0.384	0.402	0.313	0.410	
10 years	0.118	0.095	0.152	0.262	
20 years	0.110	0.142	0.212	0.292	
Panel B. Factor-adjusted network					
1 year	0.651	0.723	0.537	0.613	0.259
5 years	0.140	0.182	0.211	0.318	
10 years	0.020	0.028	0.121	0.225	
20 years	−0.041	−0.014	0.104	0.190	

Note: Panel A uses the raw state benchmark network of Section 2 and Panel B the factor-adjusted one, in the 21 annual networks 2005–2025. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). For each target and pair of endpoint years separated by the indicated gap, the rank and Pearson correlations compare the 47 off-diagonal source cells; entries are medians over the target-pairs. Top 10% and Top 20% report mean retention of the target's 5 and 10 strongest sources. The between share decomposes the variation of the connectedness cell across the 2,256 directed links and 21 endpoints. Recomputing cells as shares of each target's off-diagonal total leaves every within-target statistic unchanged; its between share is 0.381 in the raw and 0.257 in the factor-adjusted network.

OA.5.9 Direct and common-origin parts of migration exposure

Equation (8) splits exposure into residents of the target who move to the source, residents of the target who came from the source weighted by the share of the source's residents who stay, and the origins the two share. Table OA.33 reports the three parts as shares of off-diagonal exposure at each endpoint. In the state matrices, which keep stayers on the diagonal, the two direct parts account for 48.2 and 50.5 percent on average and the common-origin part for 1.3 percent, between 1.2 and 1.4 at every endpoint, and conditional exposure built from the direct parts alone correlates at 0.99998 with the full measure. The metropolitan matrices hold movers only, so an area's diagonal counts moves among its own counties and the common-origin part is 39.1 percent across all pairs and 42.8 percent across pairs in different states.

Table OA.34 repeats the specification sequence of Table 4 with the direct part alone and with the direct and common-origin parts entered together, each built as conditional exposure is built, so that the two add up to it. Among metropolitan pairs in different states, beyond the national cycle, the direct part carries the association in column (4), 0.064 (0.029), and the common-origin part is 0.013 (0.050). Across states the two parts cannot be told apart. With both entered, the raw geography slope loads on the common-origin part, 15.1 (5.6), whose standard deviation is 0.025 percentage points, but that coefficient is −9.2 in column (4) and 4.4 in column (5), and in the factor-adjusted network neither part is distinguishable from zero once geography enters.

Table OA.33: Direct and common-origin parts of migration exposure, states and metropolitan areas, 2005–2025

Endpoint	States			Metropolitan areas		
	Target's residents (D1)	Source's residents (D2)	Common origins (C)	Target's residents (D1)	Source's residents (D2)	Common origins (C)
2005	47.10	51.48	1.42	23.99	35.25	40.76
2006	47.01	51.58	1.41	24.14	35.27	40.59
2007	47.18	51.42	1.40	24.43	35.22	40.35
2008	47.59	51.03	1.38	24.82	35.16	40.02
2009	48.03	50.60	1.37	25.13	35.20	39.67
2010	48.37	50.29	1.34	25.44	35.30	39.27
2011	48.66	50.01	1.33	25.73	35.35	38.93
2012	48.67	50.02	1.31	25.78	35.58	38.63
2013	48.57	50.14	1.29	25.67	35.98	38.35
2014	48.44	50.29	1.27	25.58	36.23	38.19
2015	48.30	50.43	1.26	25.41	36.28	38.31
2016	48.13	50.60	1.27	25.20	36.20	38.59
2017	47.99	50.74	1.28	25.01	35.95	39.04
2018	48.00	50.72	1.28	25.05	35.56	39.40
2019	48.10	50.63	1.27	25.07	35.38	39.54
2020	48.31	50.46	1.23	25.23	35.53	39.24
2021	48.50	50.30	1.19	25.43	35.71	38.86
2022	48.75	50.05	1.20	25.64	35.83	38.53
2023	48.87	49.91	1.22	25.67	35.86	38.47
2024	49.04	49.73	1.23	25.86	35.91	38.22
2025	49.09	49.72	1.19	25.97	35.89	38.14
Mean across endpoints	48.22	50.48	1.29	25.25	35.65	39.10
out-of-state pairs				25.17	32.05	42.77
Mean external openness (percent)		5.69			70.51	
Correlation of $x_{ij,t}$ with its direct part		0.99998			0.96926	

Note: Shares, in percent, of total off-diagonal migration exposure $\sum_{i \neq j} \psi_{ij,t}$ in each annual network, with $\Psi_t = \Phi_t M_t$ built from IRS flows pooled over $t-15$ through $t-11$ (equation 7, Section 5.1). For target i and source $j \neq i$ exposure splits exactly into three parts, $\psi_{ij} = \varphi^{i \leftarrow i}\mu^{i \to j} + \varphi^{i \leftarrow j}\mu^{j \to j} + \sum_{k \neq i,j} \varphi^{i \leftarrow k}\mu^{k \to j}$. D1 is the target's own residents who move to the source, D2 the target's residents who came from the source weighted by the source's diagonal share, and C the common-origin part, residents who came from third units that also send migrants to the source. D1 and D2 are the direct part. States are the 48 contiguous states, whose flow matrix keeps the IRS same-state non-migrants on the diagonal. Metropolitan areas are the 100 areas of the metropolitan system, with IRS county-to-county flows aggregated to CBSA pairs and normalized within the system. That flow matrix holds movers only, so an area's diagonal is the moves between its own counties rather than its stayers, and external openness is correspondingly large. The out-of-state row restricts the metropolitan sums to pairs of areas with different principal states. The last row is the pooled correlation between conditional migration exposure $x_{ij,t}$ and the same construction applied to the direct part alone, across the 47,376 state and 207,900 metropolitan directed pair-years.

Table OA.34: The allocation ladder with the direct and common-origin parts of migration exposure, raw and factor-adjusted networks, 2005–2025

Dependent variable	Conditional connectedness share (percentage points)				
	Bivariate	Geography slope	Geography and source-by-year effects	Geography and node-year effects	Node-year effects without geography
	(1)	(2)	(3)	(4)	(5)
Panel A. Raw network					
States					
Direct part alone	0.103***	0.104**	−0.001	−0.014	0.068***
	(0.027)	(0.044)	(0.022)	(0.020)	(0.014)
	[0.346]	[0.349]	[−0.003]	[−0.046]	[0.228]
Direct part, both entered	0.045**	−0.001	0.007	−0.001	0.059***
	(0.019)	(0.033)	(0.024)	(0.022)	(0.016)
	[0.150]	[−0.003]	[0.022]	[−0.003]	[0.198]
Common-origin part, both entered	12.02**	15.09***	−4.55	−9.16**	4.38
	(5.46)	(5.59)	(4.23)	(3.83)	(5.64)
	[0.302]	[0.379]	[−0.114]	[−0.230]	[0.110]
Metropolitan areas, out-of-state pairs					
Direct part alone	0.049*	0.035	0.017	0.010	0.046***
	(0.025)	(0.026)	(0.011)	(0.010)	(0.013)
	[0.072]	[0.052]	[0.025]	[0.014]	[0.067]
Direct part, both entered	0.007	−0.005	0.016	0.010	0.036***
	(0.019)	(0.019)	(0.011)	(0.011)	(0.012)
	[0.010]	[−0.008]	[0.023]	[0.014]	[0.052]
Common-origin part, both entered	0.263***	0.261***	0.038	−0.005	0.123**
	(0.061)	(0.065)	(0.039)	(0.040)	(0.051)
	[0.130]	[0.129]	[0.019]	[−0.003]	[0.061]
Metropolitan areas, all pairs					
Direct part alone	0.170***	0.065**	0.067**	0.060**	0.192***
	(0.042)	(0.029)	(0.025)	(0.025)	(0.038)
	[0.361]	[0.138]	[0.141]	[0.128]	[0.408]
Direct part, both entered	0.127***	0.041	0.063**	0.058**	0.159***
	(0.039)	(0.028)	(0.025)	(0.025)	(0.036)
	[0.270]	[0.086]	[0.133]	[0.123]	[0.337]
Common-origin part, both entered	0.314***	0.220***	0.102***	0.075**	0.363***
	(0.052)	(0.041)	(0.036)	(0.035)	(0.067)
	[0.212]	[0.149]	[0.069]	[0.051]	[0.245]

Note: The 21 annual benchmark networks 2005–2025, raw in this part and factor-adjusted in its continuation. The dependent variable and samples are those of Table 4, whose exposure cells the two parts replace. The direct part and the common-origin part of conditional migration exposure are built as the regressor is, each divided by the target's openness, multiplied by 100 and centered within target and year, so the two add up to conditional migration exposure exactly (the parts are defined in Online Appendix Table OA.33). Each block reports the direct part entered alone and then the two parts entered together. Geography enters in columns (2) to (4) and its coefficients are not shown. Columns follow Table 4. Column (1) is the pooled bivariate slope and column (2) the geography slope, with the geography controls of that table: log distance, contiguity and same Census region with their pairwise interactions and the interactions of log distance with each indicator across states, and log distance, shared Census division and same Census region with the interactions of log distance with each indicator across metropolitan areas. Column (3) adds source-by-year effects, column (4) target-by-year effects as well, and column (5) has source-by-year and target-by-year effects without geography. Brackets contain standardized coefficients, the coefficient times the standard deviation of the part in the estimation sample, in percentage points per standard deviation. Those standard deviations are 3.345 (direct) and 0.025 (common-origin) across states, 1.457 and 0.494 across out-of-state metropolitan pairs and 2.121 and 0.676 across all metropolitan pairs. The state common-origin coefficients are printed to two decimals because that part varies little. Parentheses contain shared-node standard errors, which allow two directed pairs to be dependent whenever they share a node in either role. *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against a Student t distribution with one degree of freedom fewer than the number of nodes. The regressions run in Stata (reghdfe, with the shared-node covariance in Mata).

Table OA.34 (continued). Factor-adjusted networks

Dependent variable	Conditional connectedness share (percentage points)				
	Bivariate	Geography slope	Geography and source-by-year effects	Geography and node-year effects	Node-year effects without geography
	(1)	(2)	(3)	(4)	(5)
Panel B. Factor-adjusted network					
States					
Direct part alone	0.059^{***}	0.039	0.006	0.000	0.059^{***}
	(0.019)	(0.025)	(0.019)	(0.016)	(0.017)
	[0.196]	[0.131]	[0.022]	[0.000]	[0.197]
Direct part, both entered	0.048^{***}	0.006	0.008	0.004	0.051^{***}
	(0.016)	(0.023)	(0.022)	(0.020)	(0.015)
	[0.159]	[0.020]	[0.026]	[0.014]	[0.171]
Common-origin part, both entered	2.27	4.73	−0.78	−2.97	3.83
	(3.41)	(3.88)	(4.00)	(3.95)	(3.49)
	[0.057]	[0.119]	[−0.020]	[−0.075]	[0.096]
Metropolitan areas, out-of-state pairs					
Direct part alone	0.086^{***}	0.074^{***}	0.074^{**}	0.065^{**}	0.086^{***}
	(0.026)	(0.026)	(0.030)	(0.028)	(0.028)
	[0.126]	[0.108]	[0.107]	[0.094]	[0.125]
Direct part, both entered	0.068^{**}	0.055^{*}	0.070^{**}	0.064^{**}	0.082^{***}
	(0.028)	(0.028)	(0.030)	(0.029)	(0.030)
	[0.099]	[0.080]	[0.102]	[0.093]	[0.120]
Common-origin part, both entered	0.115^{***}	0.127^{***}	0.073	0.013	0.040
	(0.042)	(0.043)	(0.057)	(0.050)	(0.056)
	[0.057]	[0.063]	[0.036]	[0.007]	[0.020]
Metropolitan areas, all pairs					
Direct part alone	0.394^{***}	0.166^{***}	0.190^{***}	0.178^{***}	0.451^{***}
	(0.082)	(0.057)	(0.063)	(0.062)	(0.084)
	[0.836]	[0.351]	[0.403]	[0.377]	[0.957]
Direct part, both entered	0.349^{***}	0.150^{**}	0.184^{***}	0.174^{***}	0.390^{***}
	(0.082)	(0.059)	(0.064)	(0.062)	(0.081)
	[0.741]	[0.318]	[0.389]	[0.369]	[0.827]
Common-origin part, both entered	0.328^{***}	0.140^{**}	0.184^{**}	0.138^{*}	0.661^{***}
	(0.098)	(0.055)	(0.074)	(0.072)	(0.139)
	[0.222]	[0.095]	[0.124]	[0.093]	[0.447]

Note: Factor-adjusted benchmark networks. Samples, columns, regressors, standardization and inference are those of the first part of the table. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3).

OA.5.10 Pairs of areas close to each other

Schubert (2024) excludes links within 50, 100 and 150 miles of each commuting zone. Table OA.35 applies the same exclusion to the out-of-state metropolitan allocation of Table 4, dropping both directions of every pair whose internal points lie within the radius and keeping exposure as built. At 50 miles, which drops six directed pairs, the factor-adjusted column (4) estimate is 0.063 (0.026). At 100 miles, which drops 74, it is 0.047 (0.025), with an interval of $[-0.002, 0.095]$, and at 150 miles, which drops 188, 0.045 (0.028). The raw estimate falls from 0.009 to 0.002 at 100 miles. The factor-adjusted estimates of columns (1) to (3) keep intervals that exclude zero at every radius, as do all estimates across all metropolitan pairs. Most of the association that survives node-year effects therefore remains without the nearby out-of-state pairs, but it is no longer distinguishable from zero.

Table OA.35: Metropolitan allocation without nearby pairs of areas, raw and factor-adjusted networks, 2005–2025

Dependent variable	Conditional connectedness share (percentage points)					Directed
	Bivariate	Geography slope	Geography and source-by-year effects	Geography and node-year effects	Node-year effects without geography	pairs dropped
Exclusion radius	(1)	(2)	(3)	(4)	(5)	
Panel A. Out-of-state pairs, raw network						
No exclusion	0.056**	0.047*	0.018	0.009	0.046***	0
	(0.025)	(0.027)	(0.011)	(0.010)	(0.012)	
Within 50 miles	0.058**	0.049*	0.019*	0.009	0.047***	6
	(0.025)	(0.026)	(0.011)	(0.010)	(0.012)	
Within 100 miles	0.065**	0.058*	0.016	0.002	0.043***	74
	(0.030)	(0.030)	(0.014)	(0.014)	(0.016)	
Within 150 miles	0.070**	0.064**	0.016	−0.001	0.037**	188
	(0.031)	(0.031)	(0.017)	(0.018)	(0.018)	
Panel B. Out-of-state pairs, factor-adjusted network						
No exclusion	0.077***	0.069***	0.070**	0.060**	0.077***	0
	(0.021)	(0.022)	(0.027)	(0.025)	(0.024)	
Within 50 miles	0.079***	0.071***	0.074***	0.063**	0.080***	6
	(0.021)	(0.022)	(0.027)	(0.026)	(0.024)	
Within 100 miles	0.066***	0.065***	0.065**	0.047*	0.055**	74
	(0.019)	(0.020)	(0.029)	(0.025)	(0.024)	
Within 150 miles	0.065***	0.066***	0.066**	0.045	0.047*	188
	(0.020)	(0.020)	(0.030)	(0.028)	(0.027)	
Panel C. All pairs, raw network						
No exclusion	0.160***	0.071**	0.067***	0.060***	0.185***	0
	(0.036)	(0.027)	(0.023)	(0.022)	(0.033)	
Within 50 miles	0.159***	0.075**	0.070***	0.063**	0.186***	32
	(0.037)	(0.029)	(0.025)	(0.024)	(0.035)	
Within 100 miles	0.169***	0.103***	0.103***	0.092***	0.209***	194
	(0.036)	(0.030)	(0.025)	(0.025)	(0.033)	
Within 150 miles	0.151***	0.103***	0.090***	0.072***	0.184***	386
	(0.034)	(0.029)	(0.023)	(0.023)	(0.032)	
Panel D. All pairs, factor-adjusted network						
No exclusion	0.346***	0.148***	0.184***	0.171***	0.425***	0
	(0.066)	(0.046)	(0.057)	(0.056)	(0.072)	
Within 50 miles	0.338***	0.158***	0.200***	0.187***	0.424***	32
	(0.067)	(0.049)	(0.064)	(0.063)	(0.076)	
Within 100 miles	0.329***	0.198***	0.274***	0.256***	0.452***	194
	(0.062)	(0.049)	(0.066)	(0.068)	(0.074)	
Within 150 miles	0.265***	0.176***	0.245***	0.217***	0.387***	386
	(0.051)	(0.037)	(0.053)	(0.052)	(0.067)	

Note: The 100 metropolitan areas in the 21 annual networks 2005–2025. Each cell is the coefficient on conditional migration exposure in the regressions of Table 4, with exposure kept as built (not re-centered on the reduced sample). Rows drop every directed pair whose Census Gazetteer internal points lie within 50, 100 or 150 miles (80.47, 160.93 and 241.40 kilometers) of each other, the distance the geography controls use, in both directions. The last column counts the directed pairs dropped, of the 9,536 out-of-state directed pairs and the 9,900 directed pairs in all in each annual network. Out-of-state pairs exclude dyads whose members share a principal state. Columns follow Table 4. Column (2) adds its geography controls, log distance, shared Census division and same Census region with the interactions of log distance with each indicator, column (3) source-by-year effects, column (4) target-by-year effects as well, and column (5) has source-by-year and target-by-year effects without geography. In the factor-adjusted network at 100 miles, column (4) of the out-of-state pairs is 0.047 with a 95-percent interval from −0.002 to 0.095, the estimate plus or minus the 0.975 quantile of t_{99} times its standard error. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Parentheses contain shared-node standard errors, which allow two directed pairs to be dependent whenever they share a node in either role. *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against a Student t distribution with one degree of freedom fewer than the number of nodes. The regressions run in Stata (reghdfe, with the shared-node covariance in Mata).

OA.6 Measurement and Influence

OA.6.1 Alternative aggregations of the constraint measures

Table OA.36 varies the aggregation of both supply measures on the same 48-state sample and the same model, in the raw network (columns 1–3) and the factor-adjusted one (columns 4–6). Panel A varies the aggregation of WRLURI 2006, where survey weights and jurisdiction land area change the relative weights of the responding jurisdictions. A series in the aggregation inputs labeled as weighted by metropolitan population is the equal-jurisdiction mean (Appendix A.4) and is not listed. Panel B compares the four CBSA slope measures defined in Appendix A.4 with the county slope proxy and the two boundary-allocation checks, and Panel C reports the regulation coefficient under each of them. In the factor-adjusted network the TO association is 7.8 (2.2) with survey weights, 7.9 (2.4) with land-area weights and between 7.45 and 7.52 across the land measures. The regulation association remains positive under every weighting rule. No physical-constraint estimate is distinguishable from zero at the 5-percent level; improved geographic alignment does not uncover a precise positive association in this comparison.

Table OA.36: Regulation and the steep land share under alternative state aggregations, raw and factor-adjusted networks, 1985–2025

	Raw network			Factor-adjusted network		
	log TO (1)	log FROM (2)	log NET (3)	log TO (4)	log FROM (5)	log NET (6)
Panel A. Coefficient on WRLURI 2006 under each aggregation of the regulation index						
Equal jurisdictions (Table 2)	10.54**	2.76***	7.78**	7.50***	1.67	5.83**
	(3.62)	(0.60)	(3.40)	(2.32)	(1.05)	(2.08)
Survey weights	10.90**	2.72***	8.18**	7.78***	1.64	6.14**
	(3.44)	(0.59)	(3.22)	(2.18)	(1.05)	(2.06)
Jurisdiction land-area weights	9.74**	2.49***	7.25*	7.89***	1.15	6.74**
	(3.90)	(0.64)	(3.68)	(2.42)	(1.17)	(2.21)
Panel B. Coefficient on the steep land share under each aggregation						
CBSA housing stock, 1990 (Table 2)	1.19	0.72	0.47	−0.71	−0.19	−0.51
	(3.43)	(0.46)	(3.25)	(2.39)	(0.87)	(2.28)
CBSA housing stock, 2000	1.35	0.75	0.61	−0.60	−0.17	−0.43
	(3.46)	(0.47)	(3.28)	(2.42)	(0.88)	(2.30)
CBSA population, 1990	1.12	0.69	0.43	−0.76	−0.24	−0.52
	(3.37)	(0.46)	(3.20)	(2.37)	(0.87)	(2.28)
CBSA population, 2000	1.32	0.72	0.60	−0.64	−0.21	−0.43
	(3.42)	(0.46)	(3.24)	(2.41)	(0.88)	(2.31)
County slope comparator	1.60	0.83*	0.77	−0.92	0.04	−0.95
	(3.60)	(0.48)	(3.38)	(2.32)	(0.86)	(2.09)
Montana allocation sensitivity	1.19	0.72	0.47	−0.71	−0.19	−0.51
	(3.43)	(0.46)	(3.25)	(2.39)	(0.87)	(2.28)
Colorado land-area sensitivity	1.19	0.72	0.47	−0.71	−0.19	−0.51
	(3.43)	(0.46)	(3.25)	(2.39)	(0.87)	(2.28)
Panel C. Coefficient on WRLURI 2006 across the aggregations of Panel B						
Range	10.52–10.63	2.75–2.78	7.77–7.84	7.45–7.52	1.66–1.72	5.74–5.84

Note: 48 states and 41 annual networks 1985–2025 (1,968 state-years), benchmark network. Columns (1) to (3) use the raw network and columns (4) to (6) the factor-adjusted one. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Each entry comes from the regression of Table 2 with the regulation index (Panel A) or the steep land share (Panel B) replaced by the named aggregation, standardized across the 48 states. Panel C gives the range of the WRLURI 2006 coefficient over the seven regressions of Panel B. Appendix A.4 defines the aggregations. A further WRLURI 2006 series in the aggregation inputs, labeled as weighted by 2000 Census metropolitan-area population, equals the equal-jurisdiction mean stored at single precision in every state. It is therefore not an independent aggregation and is not reported. $\log \mathrm{NET} = \log(\mathrm{TO}/\mathrm{FROM})$, so its coefficients equal the TO minus the FROM coefficients exactly. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome, and on a logged control it is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws).

OA.6.2 What the regulation index measures

Table OA.37 reports the decomposition of the index into survey-content and state-rating blocks of Section 6 at the metropolitan scale, for the receiving side, in the raw and the factor-adjusted networks. The index's own component weights are recovered by regressing it on its standardized components, which returns an R^2 of one and therefore the exact weighting, and each block of components enters with those weights, so that the blocks of a column add up to the published index before each is standardized across the 100 metropolitan areas. The State Court Involvement Index is identical to the published Foster–Summers Judicial Profile to machine precision, and is uncorrelated with the survey's own questions about court involvement, at -0.003 and -0.031. In the raw network, column (1), with division effects, separates the nine components with survey content from the two external ratings. Survey content is associated with 0.75 percent more external connectedness per standard deviation (0.25), while the ratings carry 0.17 (0.35). Column (2) splits the survey content into the seven local land-use instruments—project approval, zoning approval, approval delay, open space, density restriction, exactions and local assembly—local political pressure and supply restrictions. Local political pressure carries 0.64 (0.24) and is the one block distinguishable from zero. Column (3) repeats the survey split within states, where the ratings are absorbed. There no single survey block is distinguishable from zero, although the three are jointly ($p = 0.007$), and entered alone, survey content, the local instruments and local political pressure are each positively associated with external connectedness. Collinearity is moderate, with a largest variance inflation factor of 2.8, and the blocks are jointly distinguishable from zero in every column, but the hypothesis that the block associations are equal is not rejected in any column, at $p = 0.20$, 0.18 and 0.11. In the factor-adjusted network the block point estimates hardly move, the standard errors roughly double, and no block remains distinguishable from zero; the ranking of the blocks is unchanged.

Table OA.38 is the state counterpart of the survey-content and state-rating decomposition, for log TO, log FROM and log NET in the raw network (columns 1–3) and the factor-adjusted one (columns 4–6). Panel A enters survey content and the ratings together, Panel B splits the survey content, and Panel C enters each block on its own. Across the 48 states the survey content and the local instruments correlate 0.98, and survey content and the ratings 0.67, which is why every block but supply restrictions is positively associated with external connectedness when it enters alone. For raw log FROM the survey-minus-ratings contrast is 3.20 (1.86), with a 95-percent interval of $[-0.86, 7.79]$ (column 2). Beyond the national cycle the survey content carries the receiving side, 3.25 (0.86) against -1.79 (1.43) for the ratings, a contrast of 5.04 (2.02) (column 5). The factor-adjusted sending side loads the other way. The ratings carry 7.63 (3.51) of log TO and 9.41 (3.32) of log NET, survey content 1.17 (2.56) and -2.08 (2.71), and the contrasts are -6.46 (5.55) and -11.49 (5.61), the second distinguishable from zero at the 10-percent level.

Table OA.37: Survey content against appended state ratings in the regulation index, metropolitan areas, raw and factor-adjusted networks, 1985–2025

	Blocks entered together			Each block alone	
	(1)	(2)	(3)	(4)	(5)
Panel A. Raw network					
Survey content (9 components)	0.75**			0.78***	0.37***
	(0.25)			(0.26)	(0.10)
Local land-use instruments (7)		0.30	0.25	0.71**	0.38***
		(0.33)	(0.13)	(0.29)	(0.11)
Local political pressure (1)		0.64**	0.19	0.77***	0.29***
		(0.24)	(0.12)	(0.22)	(0.10)
Supply restrictions (1)		−0.06	0.01	0.04	0.04
		(0.22)	(0.06)	(0.20)	(0.08)
Foster–Summers ratings (2 components)	0.17	0.15		0.38	–
	(0.35)	(0.35)		(0.36)	
Survey minus ratings	0.58				
	(0.45)				
Wald p, blocks jointly zero	0.011	0.012	0.007	–	–
Wald p, block associations equal	0.199	0.178	0.109	–	–
Within R^2	0.076	0.081	0.048	–	–
Panel B. Factor-adjusted network					
Survey content (9 components)	0.72			0.70	0.33
	(0.61)			(0.59)	(0.22)
Local land-use instruments (7)		0.40	0.19	0.68	0.30
		(0.68)	(0.28)	(0.62)	(0.25)
Local political pressure (1)		0.68	0.13	0.78*	0.22
		(0.42)	(0.17)	(0.39)	(0.14)
Supply restrictions (1)		−0.40	0.13	−0.25	0.15
		(0.52)	(0.10)	(0.52)	(0.10)
Foster–Summers ratings (2 components)	−0.09	−0.20		0.11	–
	(0.68)	(0.71)		(0.67)	
Survey minus ratings	0.81				
	(1.03)				
Wald p, blocks jointly zero	0.494	0.262	0.315	–	–
Wald p, block associations equal	0.433	0.390	0.964	–	–
Within R^2	0.034	0.039	0.010	–	–
Controls of Table D.3	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓
Census-division fixed effects	✓	✓	–	✓	–
Principal-state fixed effects	–	–	✓	–	✓
Metropolitan areas	100	100	64	100	64
Largest variance inflation factor	2.40	2.49	2.84	–	–

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025; the design of Appendix Table D.3, column 2 (columns 1, 2 and 4) or column 3 (columns 3 and 5). Panel A uses the raw metropolitan benchmark network and Panel B the factor-adjusted one. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The dependent variable is the log of external connectedness (FROM). Every column controls for the steep land share, lagged log population and the outside-node-set share. Each block applies WRLURI's own component weights to the CBSA means of its components and is standardized across the 100 areas, so that the blocks of a column add up to the published index. The Foster–Summers ratings are state-level and are absorbed by principal-state effects, so column 3 enters the survey blocks alone. Columns 4 and 5 report each block in its own regression, across all areas and within states. The survey-minus-ratings row is the difference between the two column 1 coefficients, tested on the reparametrized regression (95-percent interval $[-0.38, 1.52]$ raw and $[-1.46, 2.99]$ factor-adjusted). Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws). The Wald tests use the cluster-robust covariance of the block estimates with an F reference distribution. The regressors, and so the variance inflation factors, are the same in both networks. The state version is Appendix Table OA.38.

Table OA.38: What the regulation index measures across states, raw and factor-adjusted networks, 1985–2025

	Raw network			Factor-adjusted network		
	log TO (1)	log FROM (2)	log NET (3)	log TO (4)	log FROM (5)	log NET (6)
Panel A. Survey content and the ratings entered together						
Survey content (9)	6.41	3.00***	3.41	1.17	3.25***	−2.08
	(5.60)	(0.90)	(5.83)	(2.56)	(0.86)	(2.71)
Foster–Summers ratings (2)	5.14	−0.20	5.34	7.63*	−1.79	9.41**
	(6.47)	(1.09)	(6.68)	(3.51)	(1.43)	(3.32)
Survey minus ratings	1.26	3.20	−1.93	−6.46	5.04**	−11.49*
	(11.38)	(1.86)	(11.90)	(5.55)	(2.02)	(5.61)
	[−27.47, 25.25]	[−0.86, 7.79]	[−31.39, 22.08]	[−18.43, 6.64]	[0.42, 9.48]	[−24.40, 1.66]
Steep land share	1.09	1.02*	0.08	−1.61	0.34	−1.95
	(3.55)	(0.52)	(3.43)	(2.39)	(0.81)	(2.22)
log average distance	18.91	2.48	16.42	16.70	13.87**	2.83
	(17.74)	(3.61)	(16.89)	(13.03)	(5.34)	(11.65)
Lagged log population	12.02**	−0.26	12.29**	2.95	−3.68***	6.63**
	(4.19)	(0.63)	(4.42)	(2.23)	(0.76)	(2.35)
Wald p, blocks jointly zero	0.021	0.000	0.093	0.007	0.002	0.006
Wald p, block associations equal	0.912	0.092	0.872	0.250	0.016	0.046
Panel B. Survey content split, entered together with the ratings						
Local land-use instruments (7)	5.59	3.64***	1.96	2.09	3.26***	−1.17
	(6.41)	(0.95)	(6.72)	(3.16)	(1.12)	(3.49)
Local political pressure (1)	8.32	1.26	7.06	4.62	1.06	3.56
	(4.60)	(0.65)	(4.68)	(2.71)	(0.98)	(2.40)
Supply restrictions (1)	−2.12	−1.18	−0.94	−2.59	−0.44	−2.15
	(4.42)	(0.62)	(4.67)	(2.40)	(0.95)	(2.54)
Foster–Summers ratings (2)	1.59	−0.99	2.58	4.97	−2.20	7.17
	(6.42)	(1.20)	(6.45)	(4.23)	(1.75)	(3.84)
Wald p, blocks jointly zero	0.043	0.000	0.195	0.003	0.006	0.003
Wald p, block associations equal	0.551	0.007	0.736	0.042	0.121	0.027
Panel C. Each block in its own regression						
Survey content (9)	9.40**	2.89***	6.51*	5.60**	2.21**	3.39
	(3.54)	(0.60)	(3.44)	(2.15)	(0.81)	(2.05)
Local land-use instruments (7)	8.17*	2.81***	5.36	4.80**	2.18**	2.62
	(3.78)	(0.60)	(3.70)	(2.10)	(0.79)	(1.96)
Local political pressure (1)	10.97**	1.93**	9.04*	7.53**	1.10	6.43**
	(3.92)	(0.59)	(4.04)	(1.84)	(0.90)	(1.65)
Supply restrictions (1)	2.71	0.90	1.80	0.04	1.24	−1.20
	(4.15)	(0.71)	(4.08)	(1.98)	(0.75)	(1.94)
Foster–Summers ratings (2)	9.29*	1.74*	7.55	8.38***	0.31	8.07***
	(4.30)	(0.85)	(4.23)	(2.72)	(1.31)	(2.43)
Year fixed effects	✓	✓	✓	✓	✓	✓
State-years	1,968	1,968	1,968	1,968	1,968	1,968

Note: 48 states and 41 annual networks 1985–2025 (1,968 state-years), benchmark network. Columns (1) to (3) use the raw network and columns (4) to (6) the factor-adjusted one. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Each column is the regression of Table 2 with WRLURI 2006 replaced by blocks of its components. Each block applies the index's own component weights to the equal-jurisdiction state means of its components and is standardized across the 48 states, as in Table OA.37. The number after a block counts its components. Survey content is the local land-use instruments, local political pressure and supply restrictions. The two Foster–Summers ratings, the State Political Involvement and State Court Involvement Indexes, are external ratings of state legislatures and courts rather than survey answers. Every regression includes the steep land share, log average distance and lagged log population, and Panel C enters each block in its own regression. The survey-minus-ratings row is the difference between the two block coefficients of Panel A, tested on the reparametrized regression, and its brackets contain the 95-percent interval that inverts the wild bootstrap test. The largest variance inflation factor of a block is 2.24 in Panel A and 3.01 in Panel B, in every column. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome, and on a logged control it is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-t p-values that impose the null (9,999 draws). The Wald tests use the cluster-robust covariance of the block estimates with an $F(q, 47)$ reference distribution.

OA.6.3 Target and region influence

Regulation, physical constraint and connectedness are unevenly distributed across Census regions, so an association estimated on 48 states could rest on the contrast between a few of them. Table OA.39 gives the regional means over the 21 endpoints from 2005, and the influence exercise asks which states and which regions the estimate depends on. The regressions of Table 2, for each of its three outcomes, are estimated under the benchmark network with and without Census-region effects and under the own-lags-free network, and then re-estimated after removing each of the 48 targets in turn and each of the four Census regions. Outcomes and covariate scaling are unchanged throughout; omitting a target does not re-estimate a smaller national VAR or renormalize the surviving network.

Table OA.39: Regional means of regulation, physical constraints and connectedness

Region	States	WRLURI	Steep land (%)	Connectedness (pp)
South	16	−0.430	14.016	83.682
West	11	0.040	35.414	86.453
Northeast	9	0.785	15.575	88.645
Midwest	12	−0.562	4.713	83.859

Note: WRLURI and the steep land share are averaged equally over states within each Census region. Connectedness is the benchmark network's external share, averaged over the same states and 21 annual networks. The land measure uses the current 1990-housing CBSA aggregation; WRLURI is the unchanged jurisdiction mean.

Panels A to C of Table OA.40 use the raw network and Panels D to F the factor-adjusted one. In the raw network every single-state omission retains a positive WRLURI coefficient for every outcome in every model, and the state-omission ranges in Table OA.40 stay close to the corresponding full-sample estimates, at 2.52 to 3.09 percent for external connectedness under the benchmark. Removing an entire region moves the estimates considerably further. For external connectedness the minimum comes from omitting the South in the benchmark rows and from omitting the Midwest under the own-lags-free network, and with region effects the South's omission leaves only 0.17 percent; for log TO and log NET the region-controlled estimates are smallest when the Northeast is omitted and remain well above zero. That no single target produces the positive sign is not the same claim as every target carrying the same association. The reported intervals are ranges of point estimates, not confidence intervals and not a correction for residual spatial dependence; spatially robust precision is not established by deletion.

Table OA.40: WRLURI estimates under state and Census-region exclusions, raw and factor-adjusted networks, 1985–2025

		Omitting each state		Omitting one Census region				
	Full sample	Range	Minimum	Northeast	Midwest	South	West	Sign changes
Panel A. Raw network, log *TO*								
Benchmark	10.54	[9.05, 11.83]	LA	16.00	10.01	8.59	13.79	0/0
Census-region effects	21.06	[16.62, 23.38]	NY	14.06	23.86	21.42	22.02	0/0
Own-lags-free network	13.73	[12.72, 15.34]	ND	17.95	12.17	10.72	19.36	0/0
Panel B. Raw network, log *FROM*								
Benchmark	2.76	[2.52, 3.09]	LA	3.22	2.67	1.91	3.04	0/0
Census-region effects	2.17	[1.49, 2.79]	MD	2.93	2.29	0.17	2.72	0/0
Own-lags-free network	5.52	[4.99, 6.00]	NV	5.85	4.82	5.62	6.04	0/0
Panel C. Raw network, log *NET*								
Benchmark	7.78	[6.50, 9.07]	CO	12.78	7.34	6.68	10.75	0/0
Census-region effects	18.89	[13.83, 21.29]	NY	11.13	21.56	21.25	19.30	0/0
Own-lags-free network	8.21	[6.79, 9.44]	RI	12.11	7.36	5.11	13.32	0/0
Panel D. Factor-adjusted network, log *TO*								
Benchmark	7.50	[6.38, 8.57]	LA	11.15	7.00	6.71	10.42	0/0
Census-region effects	11.46	[9.69, 13.14]	NY	9.76	11.04	13.56	11.37	0/0
Own-lags-free network	9.14	[8.50, 10.18]	LA	10.78	8.18	8.74	10.68	0/0
Panel E. Factor-adjusted network, log *FROM*								
Benchmark	1.67	[0.92, 2.01]	LA	2.72	1.71	0.36	1.94	0/0
Census-region effects	1.92	[0.97, 2.59]	LA	2.11	2.24	0.15	2.14	0/0
Own-lags-free network	4.95	[4.39, 5.47]	NH	3.72	4.99	4.92	4.88	0/0
Panel F. Factor-adjusted network, log *NET*								
Benchmark	5.83	[4.99, 7.18]	RI	8.43	5.29	6.34	8.47	0/0
Census-region effects	9.55	[7.15, 10.73]	NY	7.65	8.81	13.40	9.23	0/0
Own-lags-free network	4.18	[3.71, 5.01]	CO	7.06	3.19	3.82	5.80	0/0

Note: 48 states and 41 annual networks 1985–2025. Each entry is 100 times the WRLURI 2006 coefficient of Table 2, approximately the percent difference in the outcome per standard deviation of WRLURI 2006, under the named network and model. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The benchmark rows are Table 2, the region-effects rows add Census-region effects to it, and the own-lags-free rows use the network whose own lags are left unpenalized. Each model is estimated on the full sample, after removing each of the 48 states in turn and after removing each Census region in turn. The range and the state named under Minimum refer to the 48 state omissions. Removing a state or a region does not re-estimate the national VAR or renormalize the surviving network, and covariate standardizations are those of the full sample. Ranges are ranges of point estimates, not confidence intervals. Sign changes count the state omissions and the region omissions whose estimate has the opposite sign to the full-sample estimate.

OA.6.4 Spatially robust inference

The state-clustered inference of the paper allows arbitrary dependence within a state over time but none across states, although regulation is spatially clustered. Table OA.41 reports two alternatives for the sending-side cells of Table 2. The first clusters the wild bootstrap by the nine Census divisions, which allows arbitrary dependence among the states of a division. The second computes Conley standard errors with a Bartlett kernel in the great-circle distance between state population centers, at cutoffs of 500, 1,000 and 1,500 kilometers, allowing correlation within a state at any lag and between nearby states at any lag. The factor-adjusted TO coefficient keeps an interval that excludes zero under every procedure, and its widest interval is the division-level one. The Conley standard errors are smaller than the state-clustered ones, because the cross-state terms they add are negative on net.

Table OA.41: Spatially robust inference for the regulation coefficient, raw and factor-adjusted networks, 1985–2025

		Clustered, wild bootstrap-t		Conley, Bartlett kernel		
	Estimate	48 states	9 divisions	500 km	1,000 km	1,500 km
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Factor-adjusted network, columns (4) and (6) of Table 2						
log TO	7.50	(2.32)	(2.08)	(2.07)	(2.09)	(2.00)
		[2.67, 12.70]	[1.15, 16.38]	[3.34, 11.66]	[3.29, 11.71]	[3.48, 11.52]
		{0.004}	{0.031}	{< 0.001}	{< 0.001}	{< 0.001}
log NET	5.83	(2.08)	(1.41)	(1.72)	(1.62)	(1.44)
		[1.40, 10.43]	[3.03, 11.87]	[2.38, 9.28]	[2.57, 9.09]	[2.94, 8.72]
		{0.012}	{0.012}	{0.001}	{< 0.001}	{< 0.001}
Panel B. Raw network, columns (1) and (3) of Table 2						
log TO	10.54	(3.62)	(3.07)	(3.18)	(2.92)	(2.69)
		[2.49, 18.06]	[0.59, 20.94]	[4.14, 16.93]	[4.66, 16.41]	[5.12, 15.95]
		{0.017}	{0.045}	{0.002}	{< 0.001}	{< 0.001}
log NET	7.78	(3.40)	(2.66)	(3.04)	(2.69)	(2.45)
		[0.15, 14.82]	[−1.61, 16.18]	[1.66, 13.90]	[2.37, 13.19]	[2.86, 12.70]
		{0.047}	{0.072}	{0.014}	{0.006}	{0.003}

Note: 48 states, 41 annual networks 1985–2025 (1,968 state-years). Each row is the WRLURI 2006 coefficient of the named column of Table 2, with the same regressors (WRLURI 2006, the steep land share, log average distance and lagged log population) and year effects. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Entries are 100 times the coefficient. Per standard deviation of WRLURI 2006, an entry is approximately the percent difference in the outcome. Parentheses contain standard errors, brackets 95-percent intervals and braces two-sided p-values. Column (2) repeats the inference of Table 2. It reports the cluster-robust standard error clustered by state, the p-value of a wild cluster bootstrap-t that imposes the null (9,999 draws of Webb's six-point weights, one per cluster) and the interval that inverts that test. Column (3) applies the same procedure with the nine Census divisions as clusters (three to eight states each), which allows arbitrary dependence among the states of a division. Columns (4) to (6) report Conley standard errors. They allow the errors to be correlated within a state across all endpoints and between two states within the cutoff at any lag, with weight $\max(0, 1 - d/\bar{d})$ on the great-circle distance d between the states' 2000 population centers and cutoff $\bar{d}$. They are computed after the year effects are partialled out and carry no small-sample factor, and their intervals and p-values use Student's t with 47 degrees of freedom. On average a state has 4.3, 13.9 and 23.6 other states within 500, 1,000 and 1,500 km. Every Conley covariance matrix of the four coefficients is positive definite, although at 1,500 km the matrix of Bartlett weights is not. In every row the division and Conley standard errors are below the state-clustered one, also when the Conley ones carry the small-sample factor of column (2), because the terms they add between states are negative on net. Part of that is mechanical, since the year effects make the 48 states' contributions to the score of WRLURI 2006 sum to zero, and that part grows with the cutoff. With nine clusters the bootstrap distribution of the t statistic is more dispersed, so the division intervals are wider than the state-clustered ones except for factor-adjusted log NET.

OA.6.5 Net positions by Census region

Table OA.42 describes the raw benchmark network's net positions by Census region and for each Northeast state. The comparison sets the paper beside Gabauer et al. (2024), whose Diebold–Yilmaz network of state house prices has the Northeast receiving shocks from the other regions. Over every stable monthly window, in the raw network the Northeast is on average a net receiver, at −6.2 percentage points over 1985–2025 and −16.6 from 2005, and New York is its largest net receiver, below zero in 91 percent of the windows and in all but one of those ending from 2005. The most regulated Northeast states, Massachusetts, Rhode Island and New Hampshire, are net sources on average. The regulation gradient of Section 6 is estimated among states within the same broad region as well as across regions (Section 7.1).

Table OA.42: Net position by Census region and Northeast state, raw benchmark network

		1985–2025		2005–2025	
	WRLURI 2006	Mean	Below zero (%)	Mean	Below zero (%)
Census regions					
Northeast		−6.2	57.7	−16.6	69.0
Midwest		−6.2	64.2	−6.9	65.1
South		2.6	53.5	2.6	55.8
West		8.0	51.2	17.4	49.6
Northeast states, by WRLURI 2006					
MA	2.39	7.7	46.0	12.0	46.4
RI	2.32	10.7	29.6	7.8	36.8
NH	2.11	18.2	45.2	13.0	61.2
NJ	1.43	−2.3	47.7	−26.7	76.4
ME	1.11	−11.7	70.4	−10.5	64.8
CT	0.71	16.5	30.4	9.0	44.8
PA	0.70	−21.9	71.0	−47.1	100.0
VT	0.66	−32.7	87.9	−45.1	90.8
NY	0.15	−40.1	90.6	−62.1	99.6

Note: Raw benchmark network of the 48 states, every dynamically stable 120-month window ending from July 1985 through December 2025, and those ending from January 2005. The net difference is TO minus FROM in percentage points of forecast-error variance, and a state with a negative value is a net receiver. Mean is the mean net difference and Below zero the percentage of windows in which it is negative. Regional rows average over the state-months of the region, and their percentages are of state-months. WRLURI 2006 is standardized across the 48 states. The table describes the raw network only, the network closest in design to that of Gabauer et al. (2024), and involves no second-stage regression.